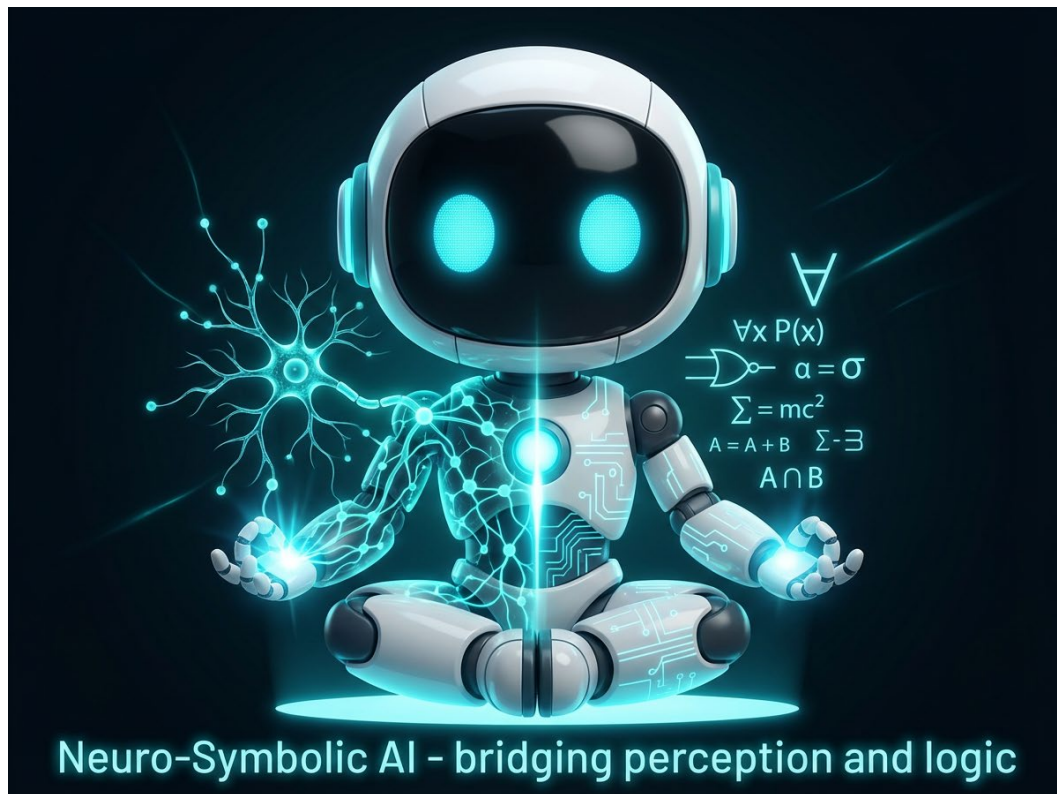


ITAI 4374: Neuroscience as a Model for AI

Module 10

Neuro-Symbolic AI & Architectures

Can a machine that recognizes faces also understand why? Bridging perception and logic.



Houston City College • AI & Robotics Program

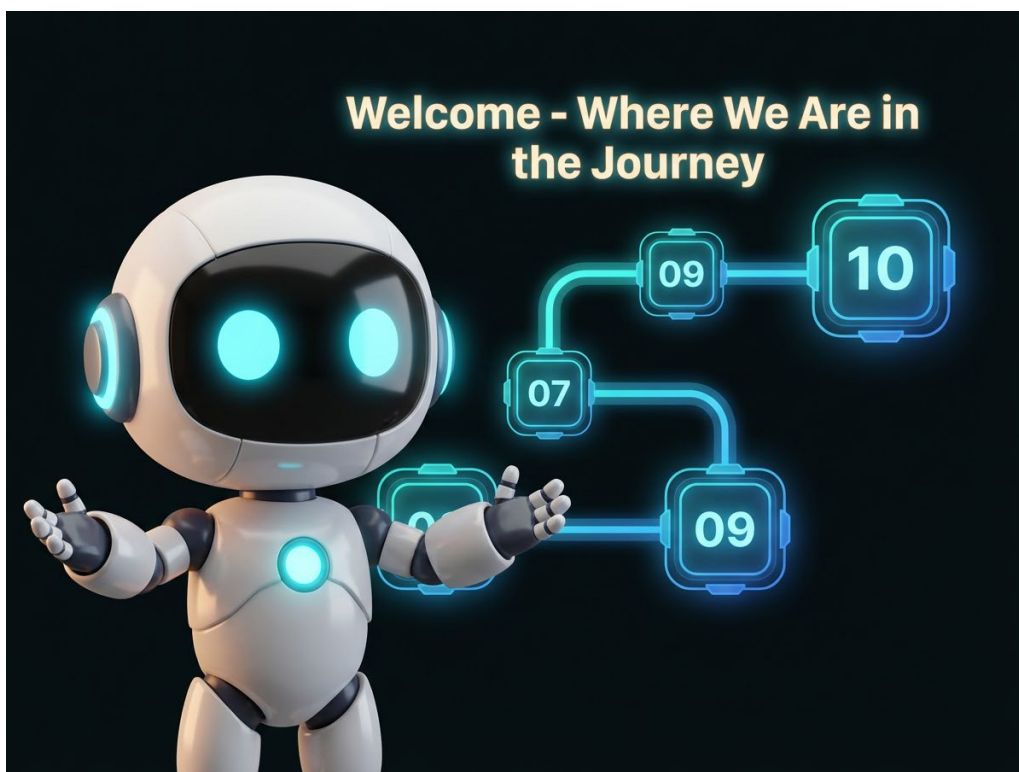
Professor Patricia McManus

Spring 2026

Section 1

Welcome — Where We Are in the Journey

Here is a challenge for you. Look at this sequence: **2, 4, 8, 16, 32, ?**. You instantly see the pattern — each number doubles. Your brain’s pattern recognition engine fires before you even consciously think about it. Now try this one: “All birds can fly. Penguins are birds. Can penguins fly?” That requires a completely different kind of thinking. You need to apply logical rules, check them against what you know, and recognize that the first statement is actually wrong. No amount of pattern recognition will help you — you need **reasoning**.



Now imagine you are building an AI system. You want it to look at a photograph and identify every object in it — that is pattern recognition, and modern neural networks are spectacular at it. But then you want the same system to answer: “If the person in this photo is holding an umbrella, and all people holding umbrellas are outside, and it is raining outside, should this person wear a raincoat?” That requires chains of logical inference, something neural networks struggle with. This module is about the quest to build AI systems that can do **both** — perceive the world like a neural network AND reason about it like a logician.

I showed you 1,000 pictures of cats and you learned to recognize cats. But can you explain WHY something is a cat? Can you apply the rule “all mammals have backbones” to a creature you have never seen before? The gap between recognizing and reasoning is one of the deepest problems in artificial intelligence — and the human brain solves it effortlessly every second of every day.

WHERE WE ARE IN THE COURSE

Module 02: History of brain science and AI — how the two fields grew together.

Module 03: Brain anatomy — the hardware of intelligence.

Module 04: Neurons and networks — how brain cells compute.

Module 05: Sensory processing — how the brain perceives the world.

Module 06: Memory and learning — how the brain stores and retrieves information.

Module 07: Attention and consciousness — how the brain selects what matters.

Module 08: Decision-making — System 1 (fast) and System 2 (slow) thinking.

Module 09: Predictive processing — the brain as a prediction machine, world models.

→ Module 10 (YOU ARE HERE): Neuro-Symbolic AI — bridging perception and reasoning.

Module 11: Embodied cognition and robotics.

Module 12: Ethics, consciousness, and the future of brain-inspired AI.

In Module 09 we built **world models** that simulate the future. Your brain constantly predicts what will happen next — the ball will fall, the car will turn, your friend will finish their sentence. But predicting **what** will happen is different from understanding **why** it happens. A weather prediction model can tell you it will rain tomorrow without understanding the physics of evaporation. Module 10 asks: how do we build AI systems that not only perceive and predict, but also **reason, explain, and generalize**?

WHAT YOU WILL LEARN IN THIS MODULE

By the end of Module 10, you will be able to:

1. Explain why neural networks excel at perception but struggle with logical reasoning, and why symbolic AI has the opposite strengths and weaknesses.
2. Describe the four major strategies for integrating neural and symbolic approaches (sequential, guided, compiled, bidirectional).
3. Connect Kahneman's System 1 / System 2 framework to neuro-symbolic AI architectures.
4. Explain concept learning — how brains and AI systems extract abstract concepts from examples.
5. Describe Global Workspace Theory and explain why it matters as a blueprint for AI architecture.
6. Identify real-world neuro-symbolic systems (AlphaFold, Google Search, DreamCoder) and explain how they combine neural and symbolic components.
7. Evaluate competing visions for Artificial General Intelligence (AGI) and articulate why neuro-symbolic integration is considered a leading approach.

💡 WHY SHOULD YOU CARE?

Every major AI lab — Google DeepMind, IBM Research, MIT CSAIL, Meta AI — has teams working on neuro-symbolic AI right now. The reason is simple: large language models like GPT-4 and Claude are astonishingly good at generating text, but they still fail at basic logic puzzles, make confident mathematical errors, and cannot reliably distinguish facts from plausible-sounding fiction.

The consensus among leading researchers is that the next breakthrough in AI will not come from making neural networks bigger. It will come from combining neural networks with structured reasoning — exactly the topic of this module. Understanding neuro-symbolic AI positions you at the frontier of the field, with career opportunities that did not exist five years ago.

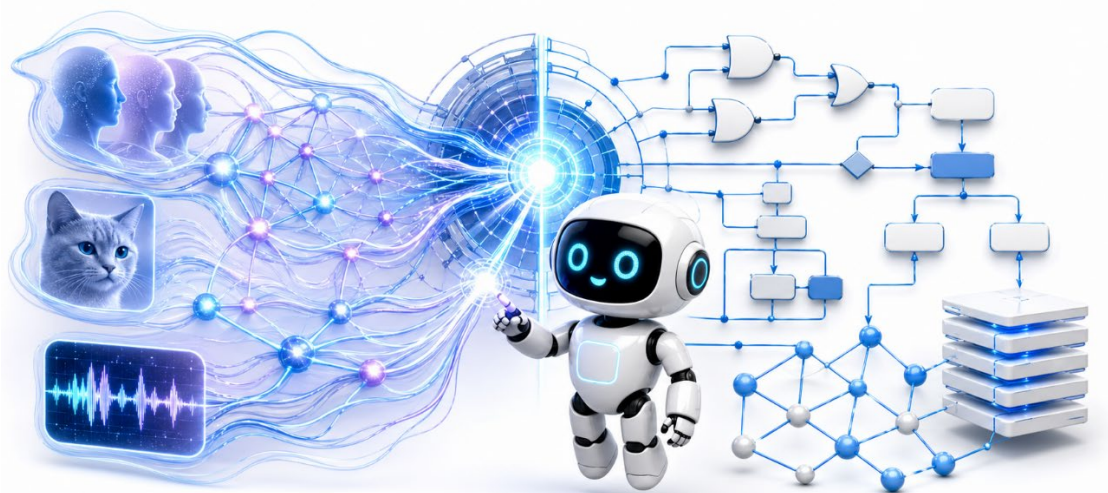
PART 1

The Two Minds of Intelligence

Neural perception meets symbolic reasoning

Section 2

Neural Networks — Amazing Pattern Recognizers, Poor Logicians



Let us start with what modern AI does incredibly well. **Neural networks** — the computational systems inspired by the brain’s web of neurons — have achieved near-human or superhuman performance on tasks that seemed impossible just fifteen years ago. They can look at a photograph and identify every object in it. They can translate between languages in real time. They can generate photorealistic images from text descriptions. They can play complex video games, beat world champions at Go, and fold protein structures that stumped scientists for decades.

Think of a neural network as an extraordinary apprentice who learns entirely by watching examples. Show it ten thousand pictures labeled “cat” and ten thousand labeled “not cat,” and it will learn to distinguish cats from non-cats with remarkable accuracy. It does this by discovering statistical patterns in the pixel data — certain textures, shapes, and spatial arrangements that reliably predict “cat.” It does not learn a definition of cat. It does not learn that cats are mammals, that they have four legs, or that they are related to lions. It learns a pattern.

This distinction matters enormously. A child who learns the concept “cat” can immediately answer questions like “is a cat bigger than a mouse?” or “could a cat survive in the ocean?” because the child understands what a cat *IS* — a living creature with certain properties that fit into a larger web of knowledge. The neural network knows only that certain pixel patterns correlate with the label “cat.”



2.1 What Neural Networks Excel At

Image recognition is perhaps the most celebrated success story. Convolutional neural networks (CNNs) can now identify objects in images more accurately than the average human — a feat demonstrated in the ImageNet competition, where error rates dropped from around 25% in 2011 to under 3% by 2015. Your phone’s ability to unlock with your face, tag friends in photos, and search your gallery by typing “beach sunset” all rely on neural networks.

Language translation underwent a revolution when neural approaches replaced older statistical methods around 2016. Google Translate, which serves billions of translation requests daily, switched to a neural architecture and immediately produced noticeably more fluent output. The transformer architecture — the same foundation behind GPT and Claude — emerged from translation research.

Game playing showcased neural networks' ability to develop strategies that surprise even experts. DeepMind's AlphaGo defeated the world champion at Go in 2016, a game with more possible board positions than atoms in the universe. The system discovered novel strategies that human players had never considered in thousands of years of play.

Content generation is the latest frontier. Models like DALL-E, Midjourney, and Stable Diffusion can create photorealistic images from text descriptions. Music generation models produce compositions indistinguishable from human work. And large language models write essays, code, and poetry that often fool expert judges. All of this is neural pattern matching at an extraordinary scale

2.2 The Brittleness Problem

Despite these triumphs, neural networks have a disturbing vulnerability: they can be **fooled by tiny changes** that would not trick a human for a second. Researchers have shown that changing a single pixel in an image — a change invisible to the human eye — can cause a neural network to confidently misclassify a cat as a toaster. These are called **adversarial examples**, and they reveal that the patterns neural networks learn are not the same patterns humans use. The network might be relying on subtle texture cues or background correlations rather than understanding what a cat actually is.

Here is a famous real-world example that illustrates the danger. Researchers trained a neural network to classify photographs as either containing military tanks or not. The system achieved impressive accuracy during testing. But when deployed in the field, it failed completely. The investigation revealed what had gone wrong: all the photographs of tanks in the training set happened to be taken on overcast days, while the non-tank photos were taken on sunny days. The network had learned to classify weather, not tanks. It had found a statistical shortcut — a correlation rather than a causal understanding.

This kind of failure is not a bug that can be fixed with more data. It reflects a fundamental limitation of the learning approach: neural networks find whatever patterns minimize their error rate, and those patterns may have nothing to do with the concepts we want them to learn. Without a way to distinguish correlation from causation, they will always be vulnerable to these shortcuts.

2.3 The Black Box Problem

Perhaps the most fundamental limitation of neural networks is that you cannot ask them **why**. When a neural network classifies an image as a cat, it cannot explain its reasoning. It cannot point to the ears, the whiskers, and the fur and say “these features indicate a cat because cats are small furry mammals with

pointed ears.” The decision emerges from millions of weighted connections in a way that is opaque even to the engineers who built the system. This is the **black box problem**.

In many applications, this opacity is acceptable. If your photo app occasionally misidentifies a dog as a cat, the consequences are trivial. But in medicine, law, finance, and autonomous driving, the inability to explain a decision is a serious problem. A doctor needs to understand why a diagnostic system recommended a particular treatment. A judge needs to understand why a risk assessment algorithm flagged a defendant. A self-driving car needs to justify why it chose to brake rather than swerve. In the European Union, the AI Act now requires that high-risk AI systems provide explanations for their decisions — a requirement that purely neural systems struggle to meet.



2.4 What Neural Networks Cannot Do

Neural networks struggle with tasks that require explicit logical reasoning, applying rules to novel situations, or generalizing from very few examples. Consider a simple logic puzzle: “Alice is taller than Bob. Bob is taller than Carol. Is Alice taller than Carol?” Humans solve this instantly through transitive reasoning — a logical rule. A neural network would need to be trained on thousands of similar examples, and even then, it would not truly “understand” transitivity. It would merely approximate the pattern.

They cannot reliably count objects in images, perform multi-step mathematical proofs, or handle scenarios that differ significantly from their training data. They struggle with negation (“show me a picture with NO cats” is surprisingly hard). They fail at compositional generalization — if they learn “red car” and “blue truck” separately, they may not correctly handle “blue car” or “red truck.” They are, in essence, extraordinarily powerful pattern matchers — but pattern matching is not the same as understanding.

AI CONNECTION: Why ChatGPT Can Write a Poem but Cannot Solve a Logic Puzzle

This is why ChatGPT can write a beautiful poem about quantum mechanics but struggles with a novel logic puzzle. The poem task is pattern matching — the model has seen millions of poems and millions of texts about quantum mechanics, and it recombines those patterns fluently. The logic puzzle requires rule application and step-by-step deduction, which are fundamentally different cognitive operations. When the model gets logic right, it is often because it has memorized a similar pattern — not because it is reasoning from first principles.

Try asking a language model to solve: “If all bloops are razzies, and all razzies are lazzies, are all bloops lazzies?” It may get this right because the pattern is simple and common in training data. But ask: “If some bloops are razzies, and all razzies are lazzies, are all bloops lazzies?” Many models fail, because the word “some” changes the logic entirely, and the model cannot reliably distinguish between these logical structures.

KEY TAKEAWAY

Neural networks are the most powerful pattern-matching systems ever built. They excel at perception — recognizing images, understanding speech, translating languages. But they learn correlations, not concepts. They cannot explain their reasoning, they are vulnerable to adversarial attacks, and they fail when asked to reason logically about novel situations. Understanding these limits is essential for knowing when to trust neural AI and when to add symbolic reasoning.

THINK ABOUT IT

Imagine you are hiring an employee. You could hire someone who has memorized the answers to every interview question ever asked (a neural network approach) or someone who understands the underlying principles and can reason about new questions they have never encountered (a symbolic approach). Which would you prefer? What if the job requires both memorization AND reasoning — like being a doctor?



Section 3

Symbolic AI — Brilliant Logician, Blind to the Real World

Now let us turn to the other side of the coin. Long before neural networks dominated headlines, AI researchers pursued a completely different approach: **symbolic AI**. Instead of learning patterns from data, symbolic AI systems reason with explicitly programmed rules and structured knowledge representations. Think of it as giving a computer a textbook of rules rather than a pile of examples.

In a symbolic AI system, knowledge is represented as **symbols** — structured pieces of information like “Cat(whiskers, four_legs, mammal)” — and **rules** — logical statements like “IF X is a mammal AND X has four legs AND X has whiskers, THEN X might be a cat.” The system reasons by applying these rules systematically, following chains of logic to reach conclusions. It is transparent, predictable, and explainable: you can trace exactly how it reached its answer.



3.1 What Symbolic AI Excels At

Chess-playing programs like IBM’s Deep Blue, which defeated world champion Garry Kasparov in 1997, were fundamentally symbolic systems. Deep Blue evaluated 200 million board positions per second using hand-crafted rules about piece values, board control, and tactical patterns. Every move could be traced back to specific rules and evaluations. When Deep Blue sacrificed a knight to gain positional advantage, engineers could trace the exact rule chain that led to that decision.

Medical diagnosis systems like MYCIN, developed at Stanford in the 1970s, used approximately 600 rules encoded by expert physicians to diagnose blood infections. MYCIN could explain every recommendation: “I concluded the patient has a Staphylococcus infection because the culture is gram-positive, the morphology is coccus, and the growth is in clusters.” In blind tests, MYCIN outperformed many practicing physicians. Its ability to explain its reasoning made it trustworthy in a way that a black-box neural network could never be.

Legal reasoning systems can navigate complex chains of precedent and statute because law is explicitly codified in rules. Tax preparation software is a form of symbolic AI — it applies the tax code’s rules to your financial data. Every deduction and credit can be traced to a specific provision of the tax code.

IBM’s **Watson**, which won the quiz show **Jeopardy!** in 2011, combined symbolic reasoning with statistical methods to parse clues, search a vast knowledge base, and formulate answers — an early hybrid system that foreshadowed the neuro-symbolic approach.

3.2 The Symbol Grounding Problem

Here is the deep philosophical challenge with symbolic AI: how does the symbol “red” connect to the actual **experience** of redness? In a symbolic system, “red” is just a label — a string of characters that can be manipulated according to rules. But for humans, “red” is a rich, multisensory concept tied to the visual experience of seeing a sunset, the warmth of fire, the taste of strawberries. This gap between abstract symbols and real-world meaning is called the **symbol grounding problem**, first articulated by philosopher Stevan Harnad in 1990.

Imagine giving a robot a dictionary written entirely in Chinese, but the robot has never seen, heard, or touched anything in the real world. It can look up any word — but every definition is written in terms of other words it has also never experienced. The robot can manipulate the symbols perfectly but understands nothing. This is the predicament of pure symbolic AI: it can reason with symbols but cannot connect those symbols to the messy, noisy, ambiguous real world.

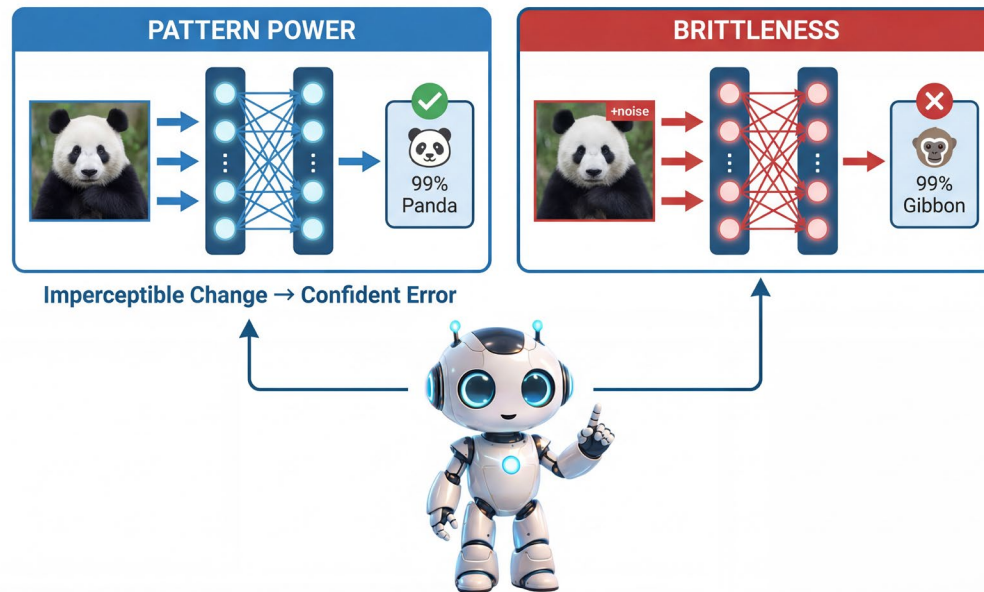
This is exactly what neural networks solve. A neural network trained on images of red objects has a grounded representation of “red” — it knows what red looks like, even if it cannot express this knowledge as a rule. The complementarity is clear: symbolic AI has concepts without grounding; neural AI has grounding without concepts.

3.3 The Brittleness of Rules

Symbolic AI is brittle in its own way — it breaks when the real world does not match the rules. Consider a medical diagnosis system with the rule: “IF patient has fever AND cough, THEN suspect influenza.” What if the patient describes their fever as “feeling hot” or their cough as “a tickle in my throat”? The symbolic system cannot handle natural language variation. What if the patient has a rare combination of symptoms not covered by any rule? The system fails silently or produces no answer at all.

The real world is messy, ambiguous, and full of exceptions. Encoding every possible situation into explicit rules is an impossible task — a challenge known as the knowledge acquisition bottleneck. This is precisely why the first wave of AI research, dominated by symbolic approaches from the 1950s through the 1980s, eventually hit a wall and gave way to the neural network revolution. The symbolic AI community had to face a humbling truth: you cannot write enough rules to capture the complexity of the real world.

NEURAL NETWORKS: PATTERN POWER AND BRITTLINESS

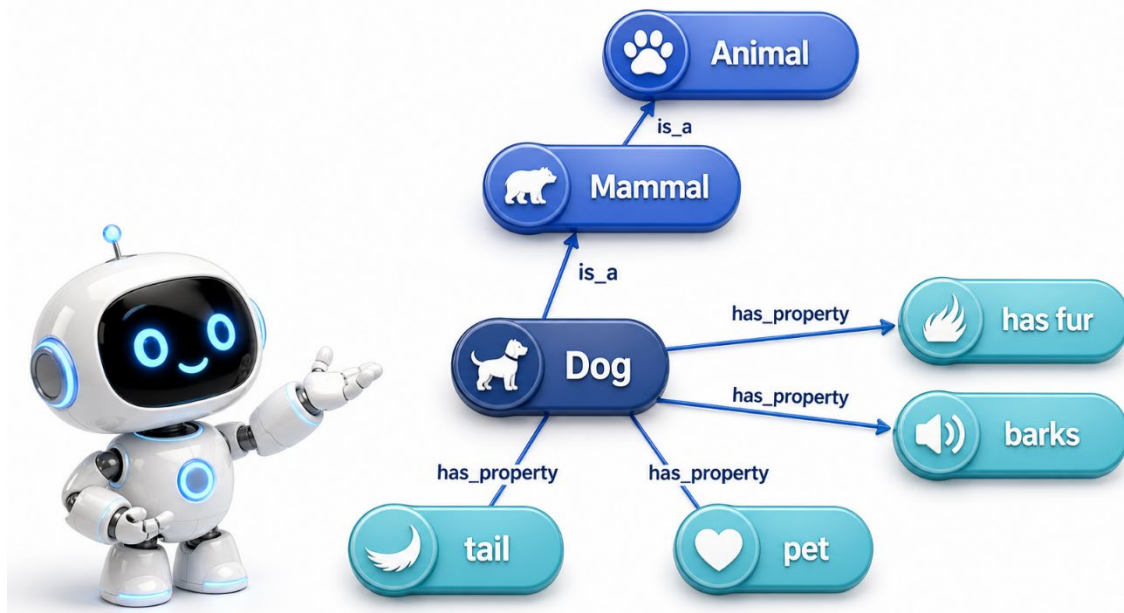


⚠ MISCONCEPTION ALERT: “Wasn’t symbolic AI proven to fail?”

No. Symbolic AI was proven to have limits — it cannot handle perception, natural language ambiguity, or situations not covered by its rules. But within its domain, symbolic reasoning is extremely powerful. It is provably correct, transparent, and explainable. The neural network revolution did not prove symbolic AI wrong — it proved that perception requires a different approach. The insight of neuro-symbolic AI is that both approaches have complementary strengths, and the future lies in combining them.

🔑 KEY TAKEAWAY

Neural networks are like an artist who can paint a perfect copy of a face but cannot tell you the person’s name or age. Symbolic AI is like a census worker who knows everyone’s name, age, and address but cannot recognize anyone by sight. True intelligence requires both the painter and the census worker working together.



Section 4

The Marriage — Why We Need Both

By now you can see the gap clearly. Neural networks are perception engines — they see, hear, and recognize patterns in the world, but they cannot explain their reasoning or apply logical rules. Symbolic AI systems are reasoning engines — they think logically, follow rules, and explain their conclusions, but they cannot perceive the messy real world or handle situations their rules do not cover. **Neither approach alone is sufficient for human-like intelligence.**

But the human brain does both, seamlessly and simultaneously. When you read this sentence, your visual cortex performs incredibly fast neural pattern recognition — converting shapes on the page into letters and words in milliseconds. At the same time, your prefrontal cortex applies grammatical rules, semantic logic, and contextual reasoning to extract meaning. The word “bank” could mean a financial institution or the side of a river, and your brain resolves this ambiguity by applying logical rules about context — a fundamentally symbolic operation — informed by the patterns recognized by your neural systems.

Consider what happens when a doctor diagnoses a patient. First, there is a **perceptual** step: the doctor looks at the patient’s skin color, listens to breathing sounds, reads X-ray images. An experienced doctor’s visual cortex has been trained on thousands of patient encounters — this is neural pattern matching. Then there is a **reasoning** step: the doctor applies medical knowledge (“these symptoms combined with that lab result suggest this diagnosis”), considers differential diagnoses, and checks their pattern-matched intuition against logical rules. This is symbolic reasoning. The interplay between perception and reasoning happens so naturally that we rarely notice it — but it is two fundamentally different computational processes working in concert.



4.1 Four Integration Strategies

Researchers have identified four major strategies for combining neural and symbolic approaches in AI systems. Each has different strengths and is suited to different applications.

Strategy 1: Sequential (Neural perceives → Symbolic reasons). The simplest approach. A neural network processes raw sensory input and converts it into symbolic representations, which are then handed to a symbolic reasoning system. For example, a computer vision system identifies objects in a security camera feed (neural), and then a rule-based system reasons about whether the observed behavior constitutes a security threat (symbolic). The neural component handles the messy perception; the symbolic component handles the clean reasoning. This is the most common integration strategy in deployed systems today.

Strategy 2: Guided (Symbolic rules constrain neural training). Instead of letting a neural network learn purely from data, you inject symbolic knowledge into the training process as constraints. For example, if you know that the output should obey the laws of physics, you can encode those laws as penalties during training. The neural network still learns from data, but it is guided to find solutions that are consistent with known rules. This is like teaching a student by giving them both examples AND textbook principles — the examples provide intuition while the principles prevent bad habits.

Strategy 3: Compiled (Distilling symbolic rules into neural weights). This approach takes explicit symbolic knowledge — such as a set of logical rules or a knowledge graph — and “compiles” it into a neural network’s weights. The resulting system runs with the speed and flexibility of a neural network but has been initialized with structured knowledge. Think of it as pre-loading a student’s brain with the textbook

before they start learning from experience. The knowledge is “baked in” to the network’s starting configuration.

Strategy 4: Bidirectional (Continuous feedback loop). The most sophisticated approach. Neural and symbolic components run simultaneously, with each continuously informing the other. The neural component provides hypotheses and perceptual interpretations; the symbolic component validates, corrects, and constrains those interpretations; the neural component updates based on the symbolic feedback. This is closest to how the human brain appears to integrate perception and reasoning, and it is the most difficult to engineer.

Strategy	How It Works	Example	Difficulty
Sequential	Neural output feeds symbolic input	Security camera + rule engine	Low
Guided	Rules constrain neural training	AlphaFold (physics rules)	Medium
Compiled	Rules baked into neural weights	Knowledge-initialized LLMs	Medium-High
Bidirectional	Continuous mutual feedback	Human brain (System 1+2)	Very High

THINK ABOUT IT

Think about how you read a novel. Your eyes scan the words (neural pattern recognition), your grammar system parses the sentence structure (symbolic rules), your semantic memory supplies word meanings (a kind of knowledge graph), and your imagination builds a mental scene (neural simulation). All of these processes are running simultaneously and feeding back into each other. Which integration strategy does this most resemble? Why might this be the hardest to build in AI?

KNOWLEDGE CHECK 1

1. What is the fundamental limitation of a neural network trained to recognize cats?

- a) It cannot process images fast enough for real-time use.
- b) It learns statistical patterns but does not understand the concept of ‘cat’ or why something is a cat.
- c) It requires too much memory to store all the cat images.

2. What is the ‘symbol grounding problem’ in symbolic AI?

- a) Symbols take too much computer memory to store efficiently.
- b) Symbolic systems are too slow for real-time applications.
- c) Abstract symbols like ‘red’ or ‘cat’ have no inherent connection to real-world experiences — they are just labels.

3. A self-driving car uses cameras to detect pedestrians (neural network) and then applies traffic rules to decide whether to stop (rule-based system). Which integration strategy is this?

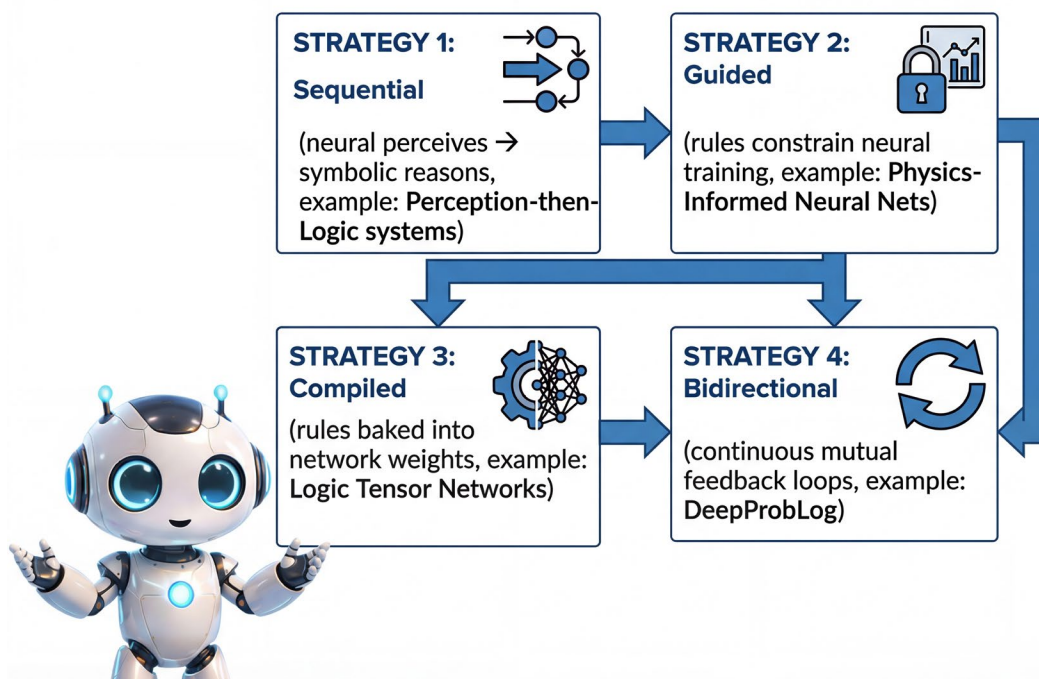
- a) Sequential: neural perceives, symbolic reasons.
- b) Guided: symbolic rules constrain neural training.
- c) Compiled: rules are embedded into neural weights.

PART 2

How the Brain Does Neuro-Symbolic Processing

The biological blueprint for hybrid intelligence

FOUR NEURO-SYMBOLIC INTEGRATION STRATEGIES



Section 5

The Dual-Process Brain — System 1 and System 2 Revisited

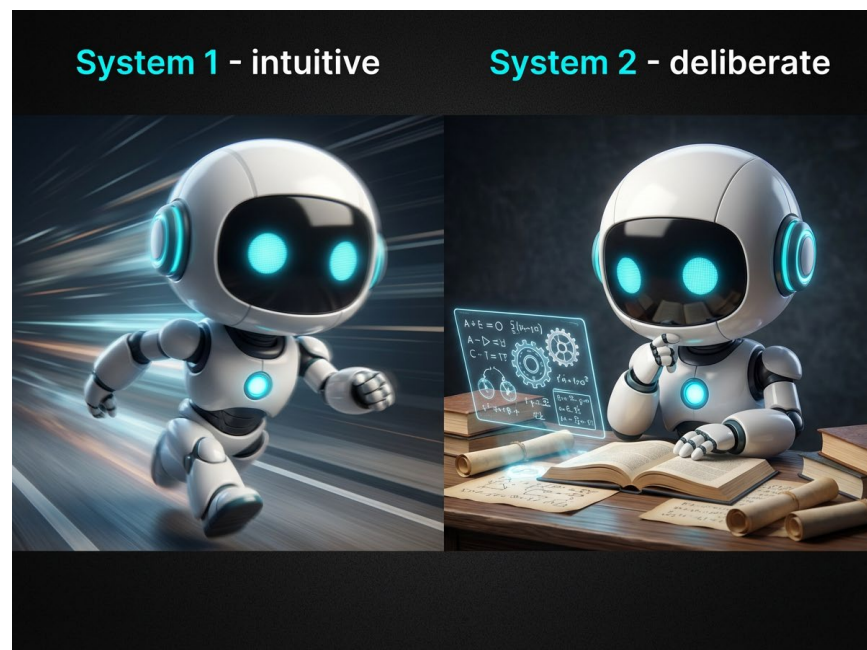
In Module 08, we explored Daniel Kahneman’s **dual-process theory** of decision-making. Kahneman described two modes of thinking: **System 1**, which is fast, automatic, and intuitive, and **System 2**, which is slow, deliberate, and logical. At the time, we focused on how these systems affect decisions. Now we are going to revisit them as an **architectural blueprint** — because System 1 and System 2 map almost perfectly onto neural and symbolic processing.

5.1 System 1: The Neural Engine

System 1 is fast, parallel, and pattern-based. When you recognize a friend’s face in a crowd, catch a ball thrown at you, or feel uneasy walking down a dark alley, System 1 is at work. It operates below conscious

awareness, processes vast amounts of sensory data simultaneously, and produces instant intuitions without explicit reasoning. You do not consciously think “that arrangement of facial features matches my stored representation of Sarah” — you simply see Sarah.

In the brain, System 1 is implemented by the **sensory cortices** (visual cortex, auditory cortex, somatosensory cortex) and the **basal ganglia** (which handle habitual, automatic responses). These areas perform massively parallel computations on incoming sensory data, extracting features, recognizing patterns, and generating rapid responses — exactly what neural networks do. System 1 also handles your emotional responses. That gut feeling about a person or situation? That is System 1 drawing on patterns from your entire life experience, processed too quickly and in too many dimensions for you to articulate consciously.



5.2 System 2: The Symbolic Engine

System 2 is slow, serial, and rule-based. When you solve a math problem, plan a vacation, or evaluate a logical argument, System 2 takes over. It requires conscious effort, can only handle one chain of reasoning at a time, and follows explicit rules step by step. System 2 is effortful — you actually **feel** it working. After an hour of difficult problem-solving, you are mentally exhausted in a way that an hour of watching television does not produce.

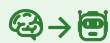
In the brain, System 2 is implemented primarily by the **prefrontal cortex** — the brain’s executive center — working in concert with **working memory**. The prefrontal cortex maintains goals, applies rules, inhibits automatic responses when they are inappropriate, and orchestrates step-by-step reasoning. It is, in effect, a biological symbolic reasoning engine. Working memory — the ability to hold a few items in mind simultaneously — provides the “scratchpad” that System 2 uses to manipulate symbols and track logical chains.

5.3 The Interplay: Hypothesize and Verify

The real power of the human brain emerges from the **interplay** between these two systems. System 1 rapidly generates hypotheses and intuitions based on pattern matching. System 2 then **evaluates**, **validates**, and if necessary **overrides** those intuitions using logical reasoning. This constant back-and-forth is the essence of intelligent behavior.

Consider a chess grandmaster deciding their next move. System 1 instantly recognizes the board pattern and suggests a move based on thousands of previous games — this is pattern matching from experience. System 2 then analytically checks: “If I move the knight here, my opponent can take my bishop, which exposes my king...” The grandmaster bounces between intuition and analysis, with each system informing the other.

Expertise, it turns out, is largely about training System 1. A **novice** doctor relies heavily on System 2 — slowly working through diagnostic criteria, checking textbook rules, consulting references. An **experienced** doctor has developed powerful System 1 pattern recognition — she walks into a room, looks at the patient, and immediately has a diagnostic intuition. But the best doctors still engage System 2 to verify their intuitions, especially in unusual cases. They have learned when to trust their instincts and when to slow down and think logically.



AI CONNECTION: System 1 and System 2 in Modern AI

Current AI systems typically implement only one of these systems. Standard neural networks are System 1 — they respond instantly to input through pattern matching, with no deliberate reasoning step. Symbolic AI systems are System 2 — they reason step by step but cannot handle raw perception.

Recent developments like “chain-of-thought prompting” in language models attempt to add System 2-like reasoning on top of System 1-like pattern matching. When you ask ChatGPT to “think step by step,” you are essentially asking it to simulate System 2 reasoning. But this simulation is fragile — the model sometimes generates plausible-looking reasoning steps that are logically invalid. True System 2 reasoning requires genuine logical operations, not just text that looks like reasoning.



KEY TAKEAWAY

System 1 (neural/pattern-matching) and System 2 (symbolic/rule-following) are not competing systems — they are complementary. The brain’s intelligence emerges from their integration. Neuro-symbolic AI attempts to replicate this integration in artificial systems.

💡 THINK ABOUT IT

Try this experiment: solve $2 + 2$ in your head. That felt effortless — System 1 handled it automatically. Now solve 347×28 . You can feel System 2 engaging — the effort, the need to hold intermediate results in working memory, the sequential steps. Now imagine trying to build an AI that can seamlessly switch between these modes. How would you design the “switch” that decides when fast intuition is sufficient and when slow deliberation is needed?

⚠️ MISCONCEPTION ALERT: “System 1 is dumb and System 2 is smart”

This is a common misunderstanding. System 1 is not “dumb” — it is incredibly sophisticated. An experienced radiologist’s System 1 can detect a tumor in an X-ray in under a second, a task that takes System 2 much longer and with less accuracy. System 1 encodes decades of experience into rapid, efficient pattern recognition. The key insight is that System 1 and System 2 are good at different things. Intelligence is knowing which system to trust in which situation.

SYSTEM 1 AND SYSTEM 2 - DUAL-PROCESS BRAIN

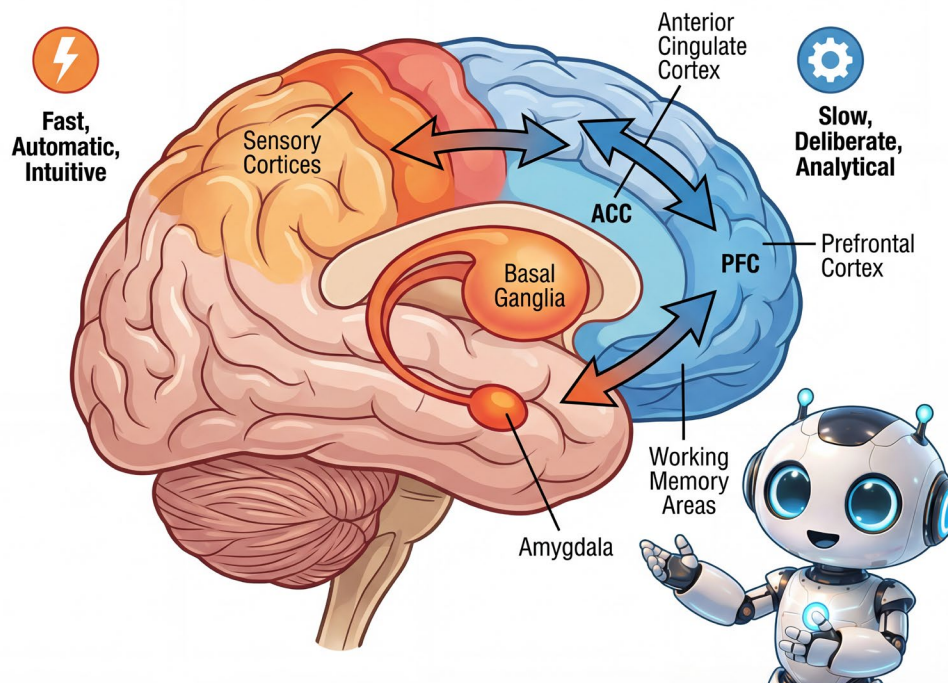


Figure 4. Conceptual overview: neural pattern recognition and symbolic reasoning as complementary modes of intelligence.

Section 6

Concept Learning — From Examples to Rules

One of the most remarkable things the human brain does is learn **concepts**. Show a two-year-old ten different chairs — a wooden dining chair, a plush armchair, a metal folding chair, a beanbag — and the child grasps the concept of “chair.” Not only can she recognize new chairs she has never seen before, but she understands what makes something a chair: it is a thing you sit on. She can even recognize chairs that look nothing like any example she has seen — a tree stump used as a chair, a stack of books arranged as a seat. The child has extracted an **abstract concept** from a handful of concrete examples.

6.1 One-Shot and Few-Shot Learning

This ability is called **few-shot learning** — learning a new concept from just a few examples — and in extreme cases, **one-shot learning** — learning from a single example. A child who sees a pangolin for the first time at the zoo can recognize pangolins in books, cartoons, and nature documentaries, even though each representation looks completely different. The child did not need ten thousand labeled pangolin images. She needed one.

Contrast this with a standard neural network, which typically needs thousands or millions of examples to learn to recognize a new category. The neural network learns statistical patterns across huge datasets. The human child extracts a structural representation — a model of what the thing IS, not just what it looks like — from minimal data. This gap between human concept learning and machine pattern learning is one of the most active areas of AI research.



Why is the human brain so much more efficient? Researchers believe it is because we do not learn concepts from scratch. Every new concept is connected to a vast web of existing knowledge. When you learn what a pangolin is, you immediately know it is an animal (it moves, it eats, it has babies), it has scales

(like a fish or a reptile), and it is terrestrial (it walks on land). This rich web of prior knowledge — a kind of mental knowledge graph — constrains and guides the learning process. This is fundamentally neuro-symbolic: neural pattern recognition informed by symbolic knowledge structures.

6.2 How the Brain Extracts Concepts

The brain builds concepts through **hierarchical representations** in the visual cortex and beyond. At the lowest levels, neurons respond to simple features — edges, colors, textures. Higher levels combine these into more complex representations — shapes, parts, objects. At the highest levels, neurons in the **inferotemporal cortex** respond to entire categories — there are neurons that fire for any face, any chair, or any tool. These high-level representations are abstract enough to generalize across very different-looking examples.

But concept learning goes beyond visual categories. Humans also form **abstract concepts** that have no direct sensory referent. “Justice,” “democracy,” “momentum” — these concepts cannot be pointed to or photographed. They are learned through language, analogy, and reasoning. The brain represents them through networks of associations in the **temporal and prefrontal cortices**, connecting them to related concepts, examples, and rules. This is inherently a neuro-symbolic process: neural representations (distributed patterns of activity) encoding symbolic structure (category relationships, logical rules).

6.3 AI Approaches to Concept Learning

Concept Bottleneck Models are a recent approach where a neural network is trained to first predict human-interpretable concepts (e.g., “has wings,” “has feathers,” “can fly”) before making a final classification (e.g., “is a bird”). This forces the network to build intermediate representations that map onto symbolic categories, creating a bridge between neural perception and symbolic reasoning. If the model classifies something as a bird, you can check whether it correctly identified the relevant features — providing transparency that standard neural networks lack.

Program synthesis takes a different approach entirely. Instead of learning to classify, the system learns to write a program that generates the observed examples. MIT’s DreamCoder system, for instance, can look at a few examples of a visual pattern and synthesize a small program that produces that pattern. The program IS the concept — a compact, interpretable, generalizable symbolic representation learned from neural-style pattern observation.

Prototype networks learn a single representative example (prototype) for each concept and classify new inputs by measuring their distance to each prototype. This mirrors how psychologists believe humans store some categories — as a mental “average” of all examples seen. When you think of “bird,” you likely imagine something like a robin or sparrow — a prototypical bird — rather than an ostrich or penguin.

 AI CONNECTION: Are GPT-4's 'Emergent Abilities' Genuine Concept Learning?

Large language models like GPT-4 display what researchers call “emergent abilities” — capabilities that appear spontaneously as the model gets larger, without being explicitly trained. GPT-4 can solve some logic puzzles, write code, and reason about novel scenarios. Is this genuine concept learning?

The debate is fierce. Skeptics argue that LLMs are doing sophisticated pattern matching on their massive training data — they have seen so many examples that they can approximate reasoning by recombining memorized patterns. Proponents argue that at sufficient scale, pattern matching becomes indistinguishable from genuine understanding. The neuro-symbolic perspective suggests that true concept learning requires the ability to extract explicit, reusable rules — something LLMs do not reliably do.

 KEY TAKEAWAY

Concept learning is the bridge between perception and reasoning. The brain builds concepts through hierarchical neural representations that become abstract enough to support symbolic reasoning. AI approaches like Concept Bottleneck Models, program synthesis (DreamCoder), and prototype networks are different attempts to bridge this same gap — converting neural patterns into structured, reusable, generalizable knowledge.

 THINK ABOUT IT

Consider the concept of “fairness.” Could a neural network learn this concept from examples alone? What kinds of examples would it need? Would the resulting concept be the same as what a symbolic system could represent using explicit rules? Which approach — neural or symbolic — would be better for a system that needs to make fair decisions about loan approvals?

? KNOWLEDGE CHECK 3

1. Which of the following best describes the difference between System 1 and System 2 thinking?

- a) System 1 is conscious; System 2 is unconscious
- b) System 1 is fast and intuitive; System 2 is slow and deliberate
- c) System 1 uses the left hemisphere; System 2 uses the right
- d) System 1 is rational; System 2 is emotional

2. A child looks at many dogs and eventually forms the general concept of "dog". Which cognitive process does this best illustrate?

- a) Pure pattern matching with no abstraction
- b) Concept learning — building abstract categories from examples
- c) Symbolic reasoning without perception
- d) Reflex conditioning

3. Why do researchers argue that concept learning is the bridge between perception and reasoning?

- a) Because concepts are purely symbolic and have nothing to do with perception
- b) Because concepts are built from perceptual data and can then be used in rule-like reasoning
- c) Because concepts exist only in language, not in the visual system
- d) Because concept learning is the same thing as memorization

Answers available from your instructor.

Section 7

The Connectome as a Knowledge Graph

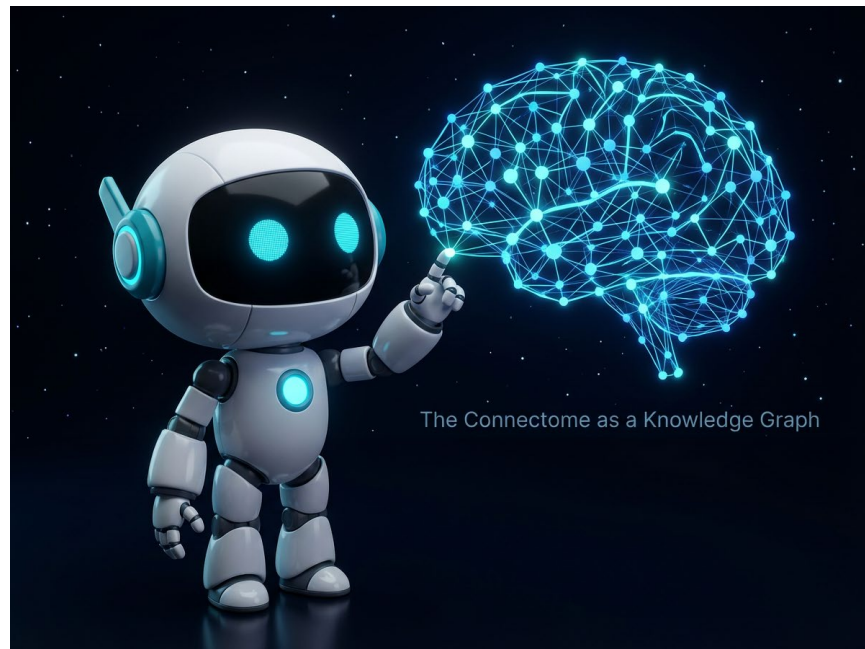
The brain's physical wiring — the **connectome** — is itself a kind of knowledge representation. The way neurons are connected to each other encodes relationships between concepts, just as a knowledge graph encodes relationships between entities. Understanding this parallel helps us see why combining neural networks with structured knowledge is such a natural idea.

7.1 Brain Wiring as Structured Knowledge

In your brain, concepts are not stored in individual neurons but in patterns of connectivity. The concept of “dog” involves connections between visual areas (what dogs look like), auditory areas (what dogs sound like), motor areas (how to pet a dog), emotional areas (how dogs make you feel), and linguistic areas (the word “dog”). These connections form a rich, multi-dimensional web of associations that gives the concept its full meaning.

Brain regions form functional circuits that process information in specific sequences. Visual information flows from the eyes to the primary visual cortex (V1), then through a hierarchy of visual areas (V2, V4, inferotemporal cortex) that extract increasingly abstract features. From there, it connects to semantic

regions in the temporal lobe (what is this?), motor regions (what can I do with it?), and emotional regions (is this good or bad?). This flow — visual → semantic → motor — mirrors the structure of knowledge graphs, where nodes represent concepts and edges represent relationships.



7.2 When Damage Reveals Structure

Some of the most compelling evidence for structured knowledge in the brain comes from patients with selective brain damage. Neurologists have documented patients who lose specific categories of knowledge while retaining others perfectly. Patient JBR, studied by Elizabeth Warrington and Tim Shallice, could identify tools and man-made objects but could not identify living things. He could recognize a hammer or a telephone but not a dog or a flower. Another patient showed the opposite pattern — intact knowledge of living things but impaired knowledge for tools.

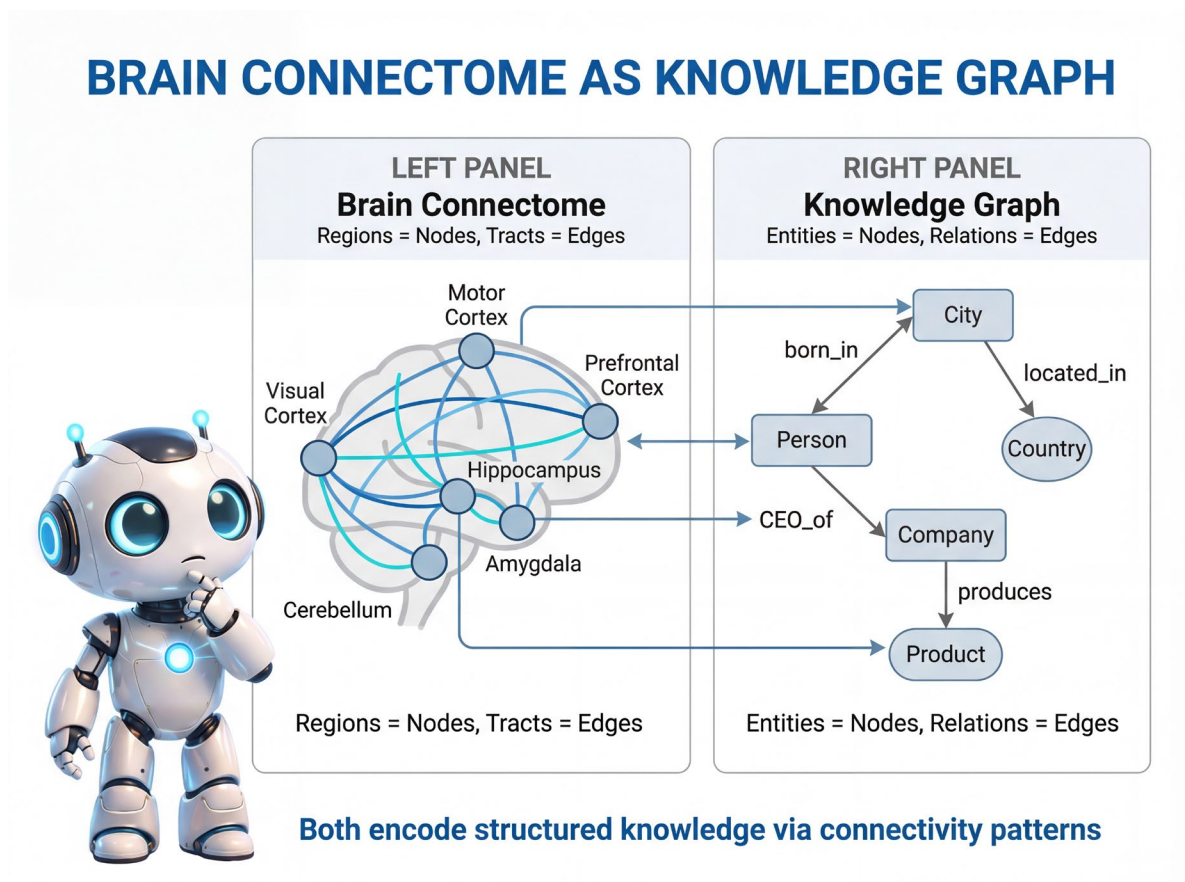
These cases suggest that the brain organizes knowledge into categories that are physically implemented in different brain regions. When a specific region is damaged, the corresponding category of knowledge is selectively lost. This is powerful evidence that the brain implements something like a knowledge graph — a structured, organized representation of concepts and their relationships — not just a homogeneous soup of neural connections.

Even more striking are patients with “category-specific” naming deficits. One famous patient could name every fruit and vegetable but could not name any animal. Another could name tools but not body parts. These highly selective deficits demonstrate that the brain’s knowledge representation has a clear structure that mirrors the categorical organization we see in knowledge graphs.

7.3 Knowledge Graphs in AI

AI researchers have built explicit knowledge graphs that mirror this kind of structured representation. **Google's Knowledge Graph**, launched in 2012, contains billions of facts about entities and their relationships — “Albert Einstein was born in Ulm,” “Ulm is a city in Germany,” “Germany is a country in Europe.” When you search for “Albert Einstein” and see a structured information panel, that data comes from the Knowledge Graph.

Wikidata, the structured knowledge base behind Wikipedia, contains over 100 million items with hundreds of millions of relationships between them. These knowledge graphs provide exactly what neural networks lack: explicit, structured, navigable representations of knowledge that can be reasoned about logically.



AI CONNECTION: Retrieval-Augmented Generation (RAG) as a Bridge

One of the most practical neuro-symbolic techniques in use today is Retrieval-Augmented Generation (RAG). Here is how it works: instead of relying solely on its neural weights (internal “memory”), the language model first searches an external knowledge base — a structured database, a document collection, or a knowledge graph — and retrieves relevant information. It then incorporates this retrieved knowledge into its response.

RAG is conceptually similar to how the hippocampus retrieves stored memories to inform current thinking. The neural network provides the language and reasoning capabilities; the external knowledge base provides structured, verifiable facts. This combination reduces hallucination (making things up) and allows the system to access knowledge that was not in its training data. If you have used an AI chatbot that can search the web before answering, you have experienced RAG in action.

KEY TAKEAWAY

The brain’s connectome is a biological knowledge graph — concepts are represented by patterns of connectivity, and categories are physically organized across brain regions. AI knowledge graphs like Wikidata and Google’s Knowledge Graph provide neural networks with the structured, explicit knowledge they otherwise lack. RAG is the most widely deployed bridge between neural and symbolic knowledge.

PART 3

Neuro-Symbolic Architectures in AI

Building systems that perceive AND reason

Section 8

Global Workspace Theory — The Theatre of Consciousness

In Module 07, we encountered Bernard Baars' **Global Workspace Theory (GWT)** as a model of consciousness. Now we are going to see why it has become one of the most influential blueprints for AI architecture — because it offers a compelling solution to the integration problem: how do you combine multiple specialized processing modules into a unified, coherent system?

8.1 The Theatre Metaphor

Baars imagined consciousness as a **theatre**. In this theatre, a bright spotlight illuminates a small stage — this is the **global workspace**, the contents of your conscious awareness at any given moment. Behind the scenes, in the darkness of the audience, sit numerous specialized processors — experts in vision, hearing, language, memory, motor planning, emotion, and more. These “audience members” work independently and in parallel, each processing their own domain of expertise.

The key mechanism is broadcasting. When a particular piece of information wins the competition for the spotlight — when it becomes the focus of attention — it is broadcast to the entire theatre. Every specialist processor receives this information simultaneously. This broadcasting creates a shared context that allows different modules to coordinate their activities. The vision module knows what the language module is processing; the memory module knows what the attention system is focused on; the motor planning module knows what the emotional system is feeling.

Think of it like a meeting room with experts from different departments. Most of the time, each expert is working independently on their specialty. But when someone stands up and makes an announcement — “We just lost our biggest client!” — every expert hears it simultaneously and can integrate this new information into their own work. The announcement is the global broadcast; the meeting room is the workspace; the experts are the specialist processors.



8.2 Brain Implementation

In the brain, the global workspace appears to be implemented by **thalamo-cortical broadcasting**. The thalamus acts as a central relay station, and long-range connections between the prefrontal cortex and other brain regions enable the broadcast mechanism. When information enters the global workspace — when it becomes conscious — it triggers a cascade of activity across widely distributed brain areas, making it available for report, memory encoding, and voluntary action.

The **prefrontal cortex** plays a special role as the “director” of the workspace — it determines what gets the spotlight, maintains information in the workspace through sustained neural activity (working memory), and coordinates the responses of different specialist modules. Damage to the prefrontal cortex impairs the ability to integrate information across domains — patients can still perceive, remember, and plan individually, but they struggle to coordinate these abilities.

8.3 GWT as an AI Architecture

Here is why GWT matters for AI: it provides a principled way to integrate multiple neural networks with a shared symbolic working memory. Imagine an AI system with:

- A **vision module** (a neural network trained on image recognition)
- A **language module** (a neural network trained on natural language processing)
- A **memory module** (a retrieval system connected to a knowledge base)
- A **reasoning module** (a symbolic logic engine)
- A **motor planning module** (for controlling actions in the world)

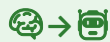
All connected through a **global workspace** — a shared symbolic representation that any module can read from or write to. The vision module detects a stop sign and writes “stop sign detected, 20 meters ahead”

to the workspace. The reasoning module reads this, consults traffic rules, and writes “vehicle must stop.” The motor planning module reads this and generates a braking action. This is a neuro-symbolic architecture: specialist neural networks communicating through a shared symbolic workspace.

8.4 Modern Implementations

You may have already encountered approximate implementations of GWT without realizing it. **LangChain** and similar agent frameworks create AI systems where a central orchestrator (the workspace) coordinates calls to specialized tools — a web search engine, a calculator, a code interpreter, a database query engine. Each tool is a specialist module; the orchestrator maintains a shared context and decides which tool to engage next.

OpenAI’s function calling, Anthropic’s tool use, and Google’s Gemini agents all implement a version of this architecture. The language model acts as the workspace, maintaining context and coordinating specialist modules. When you ask an AI assistant to “search the web for recent news about Tesla, then calculate the year-over-year revenue change, then write a summary,” you are engaging a GWT-like architecture where the central model dispatches tasks to specialized tools and integrates their results.



AI CONNECTION: The LLM Context Window as a Crude Global Workspace

The context window of a large language model — the text it can see and process at once — functions as a rudimentary global workspace. Information placed in the context is “broadcast” to all the model’s internal attention heads, which act as specialist processors. The model can combine information from different parts of the context, attending to relevant details and integrating them into coherent outputs.

But the analogy is limited. A true global workspace has persistent, structured memory that survives beyond a single processing cycle. The LLM context window is erased between conversations. A true global workspace coordinates genuinely independent specialist modules; the LLM’s “specialists” are all entangled in a single monolithic model. Building a genuine global workspace architecture for AI remains an active research challenge.

💡 THINK ABOUT IT

Right now, you are reading this booklet. Your visual system is decoding the letters (a specialist module). Your language system is parsing the grammar (another specialist). Your semantic memory is supplying the meanings of words like 'thalamus' and 'broadcast' (another specialist). And your executive system is maintaining focus on the text instead of checking your phone (the workspace director). All of this coordination happens effortlessly through your own biological global workspace. Now imagine building each of these specialists as a separate AI model and trying to coordinate them through a shared workspace. What challenges would you face?

🔑 KEY TAKEAWAY

Global Workspace Theory is more than a model of consciousness — it is a practical architectural blueprint for AI. By creating a shared workspace where specialist neural modules can broadcast information and coordinate their activities, we can build integrated systems that are more than the sum of their parts. The emergence of AI agent frameworks like LangChain represents the first generation of these architectures.

GLOBAL WORKSPACE THEORY - THEATRE OF CONSCIOUSNESS

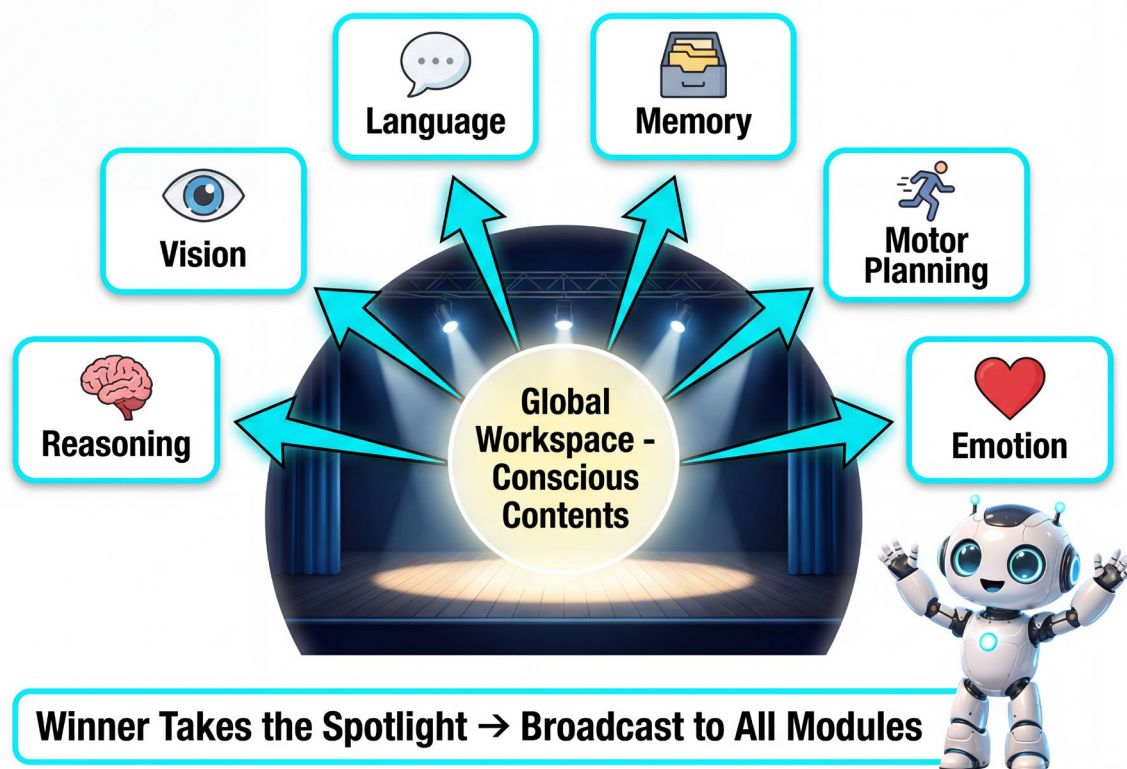


Figure 6. Architecture diagram: specialized modules communicating through a shared workspace — the biological inspiration for neuro-symbolic integration.

Section 9

Building Neuro-Symbolic Systems — Real Architectures

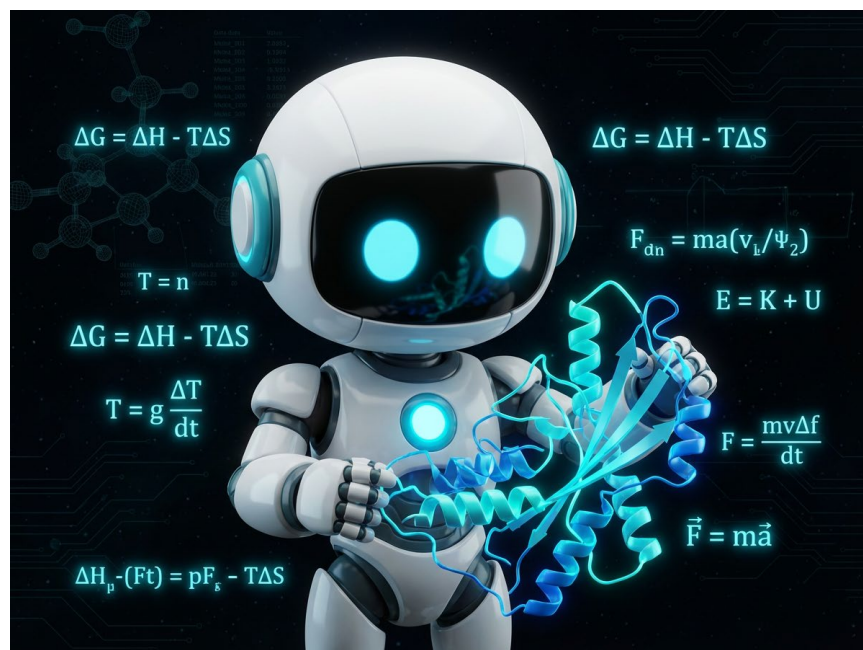
Let us look at real systems that combine neural and symbolic approaches. These examples show that neuro-symbolic AI is not just a theoretical idea — it is being deployed in some of the most important applications in the world.

9.1 Architecture 1: Neural + Physics Rules (AlphaFold)

DeepMind's **AlphaFold** solved one of biology's greatest challenges: predicting the three-dimensional structure of a protein from its amino acid sequence. This is a problem that had stumped scientists for fifty years. AlphaFold succeeded by combining deep neural networks with known biophysics rules.

The neural component learns patterns from known protein structures — how certain amino acid sequences tend to fold in particular ways. But the system does not rely on the neural network alone. It also incorporates symbolic constraints from physics and chemistry: bond lengths, bond angles, steric clashes (atoms cannot occupy the same space), and energy minimization principles. These symbolic rules guide and constrain the neural network's predictions, ensuring that the predicted structures are physically plausible. This is the “guided” integration strategy in action.

The impact has been enormous. AlphaFold has predicted structures for nearly every known protein — over 200 million structures — accelerating drug discovery and biological research by years. It won the 2024 Nobel Prize in Chemistry. And it could not have succeeded as a pure neural network or a pure symbolic system. The neural network alone would produce structures that violate physical laws. The symbolic rules alone could not navigate the astronomically large space of possible configurations. Together, they solve what neither could solve alone.



9.2 Architecture 2: Neural + Knowledge Graph (Google Search)

Google Search is perhaps the most widely used neuro-symbolic system in the world, serving billions of queries daily. The **neural component** (originally BERT, then MUM, now Gemini-based) understands the meaning of your query — it handles synonyms, ambiguity, context, and natural language nuances. The **symbolic component** — the Knowledge Graph — provides structured factual information: “Paris is the capital of France, located at coordinates 48.8566° N, 2.3522° E, with a population of 2.1 million.”

When you search for “tallest building in France,” the neural component understands what you are asking, and the Knowledge Graph provides the structured data to answer precisely. The information panel that appears beside your search results — with facts, images, and related entities neatly organized — is the Knowledge Graph in action, surfaced by neural understanding of your query.

9.3 Architecture 3: Neural + Program Synthesis (DreamCoder)

MIT’s **DreamCoder** represents one of the most creative approaches to neuro-symbolic AI. Given a few examples of a pattern or task, DreamCoder synthesizes a small program that generates those examples. It then builds a **library** of reusable program components that it can combine in new ways to solve new tasks — essentially discovering its own symbolic abstractions from neural observation.

For instance, given examples of simple drawings (squares, triangles, spirals), DreamCoder learns reusable concepts like “draw a line,” “turn 90 degrees,” “repeat N times.” It can then combine these concepts to draw shapes it has never seen before. This is concept learning in its purest form: extracting reusable, composable abstractions from examples. The learned programs are fully interpretable — you can read them and understand exactly what the system has learned.

9.4 Architecture 4: Neural + Formal Verification (Safety-Critical Systems)

In aviation, medicine, and autonomous driving, AI systems must meet stringent safety requirements. **Formal verification** — mathematically proving that a system satisfies certain safety properties — is a purely symbolic technique. Combining neural networks with formal verification allows systems to leverage the perception capabilities of neural networks while guaranteeing certain safety properties symbolically.

For example, an autonomous driving system might use neural networks for perception (detecting pedestrians, lane markings, traffic signs) but add a symbolic safety layer that enforces hard constraints: “Never accelerate when a pedestrian is detected within 10 meters.” “Never exceed the speed limit by more than 5 mph.” “Always maintain a 2-second following distance.” The symbolic layer cannot be overridden by the neural network, no matter how confident the neural network is. This creates a defense-in-depth architecture where neural flexibility is bounded by symbolic guarantees.

9.5 The Wolfram + Python Hybrid

For our lab in this module, we will use a practical neuro-symbolic hybrid that pairs **Python** (with TensorFlow/Keras for neural network components) with the **Wolfram Language** (for symbolic

computation). The Wolfram Language offers built-in capabilities for symbolic mathematics, knowledge representation, constraint solving, and natural language understanding — capabilities that would require extensive custom coding in Python alone.

This combination mirrors the neuro-symbolic pattern: Python handles the neural perception (classifying images, processing sensor data), while Wolfram handles the symbolic reasoning (applying constraints, solving equations, verifying logical consistency). Data flows from Python to Wolfram and back, with each tool doing what it does best. This is the Sequential integration strategy — the simplest and most practical starting point for building neuro-symbolic systems.

Architecture	Neural Role	Symbolic Role	Real System
Neural + Physics	Learn patterns from protein data	Enforce physics/chemistry rules	AlphaFold
Neural + Knowledge	Understand query meaning	Provide structured facts	Google Search
Neural + Programs	Observe patterns in examples	Synthesize programs as concepts	DreamCoder
Neural + Verification	Perceive environment	Enforce safety guarantees	Autonomous vehicles
Neural + Symbolic Compute	Classify images (Python/TF)	Solve constraints (Wolfram)	Lab 10

KEY TAKEAWAY

Neuro-symbolic AI is not a theoretical dream — it is deployed in some of the world's most important applications. AlphaFold won a Nobel Prize. Google Search serves billions of queries. Autonomous vehicles protect human lives. The common thread: neural perception and symbolic reasoning working together, each doing what it does best.

THINK ABOUT IT

Consider a hospital that wants to build an AI system to help diagnose diseases from medical images (X-rays, MRIs). Which of the four architectures discussed above would be most appropriate? Should the system use neural networks alone, symbolic rules alone, or a combination? What role should formal verification play in ensuring patient safety?

Section 10

The Road to Artificial General Intelligence

The question that drives much of AI research today is: how do we build **Artificial General Intelligence (AGI)** — a system with the broad, flexible, general-purpose intelligence of a human? Current AI systems are narrow specialists. GPT-4 is brilliant at language tasks but cannot drive a car. AlphaFold can predict protein structures but cannot have a conversation. A self-driving car can navigate streets but cannot write a poem. Human intelligence is general — you can learn to do all of these things and more, adapting fluidly to new challenges you have never encountered before.

10.1 Why Leading Researchers Bet on Neuro-Symbolic AI

Three of the most influential voices in AI research — **Yoshua Bengio** (Turing Award winner, University of Montreal), **Yann LeCun** (Turing Award winner, Meta AI), and **Gary Marcus** (New York University, author of “Rebooting AI”) — have all argued, from different angles, that scaling neural networks alone will not achieve AGI. Their reasoning converges on a common insight: human intelligence fundamentally involves both pattern recognition and symbolic reasoning, and any system that lacks one of these capabilities will hit a ceiling.



Bengio has specifically called for AI systems that can perform “System 2 deep learning” — neural networks that can reason step by step, handle causal relationships, and generalize out of distribution. His vision is neural networks augmented with structured reasoning capabilities, trained to discover and apply abstract rules. This is neuro-symbolic AI expressed in the language of deep learning research.

10.2 The ‘Deep Learning Plateau’ Argument

The **deep learning plateau** argument suggests that simply making neural networks larger, training them on more data, and giving them more computation will yield diminishing returns. The evidence for this includes several persistent patterns:

Large language models still make basic logical errors, even at enormous scale. They confidently assert false facts (hallucination). They fail at tasks requiring systematic generalization — applying learned rules to new combinations they have not seen. Increasing model size from billions to trillions of parameters yields smaller and smaller improvements on reasoning benchmarks. If scaling alone were sufficient, these problems should be shrinking faster than they are.

The counter-argument, advanced by some at OpenAI and Google, is that we simply have not scaled enough yet — that with ten or a hundred times more computation, these problems will dissolve. This is called the “scaling hypothesis.” The jury is still out, but the persistence of fundamental reasoning failures despite massive scaling has made many researchers skeptical.

10.3 Yann LeCun’s JEPA — An Alternative View

Yann LeCun has proposed a different path to AGI that does not rely on traditional symbolic AI. His **Joint Embedding Predictive Architecture (JEPA)** envisions a system that learns world models through self-supervised learning — predicting future states of the environment in an abstract representation space, without needing explicit symbolic rules.

LeCun argues that symbolic reasoning is an emergent property that can arise from sufficiently powerful predictive models, rather than something that needs to be explicitly programmed. In his view, the brain does not have a separate “symbolic module” — it has neural circuits that have learned to perform reasoning-like operations through experience. This is a minority view among neuro-symbolic researchers, but it comes from one of the field’s most accomplished scientists, and it deserves serious consideration.

10.4 Gary Marcus’s Case for Hybrid Approaches

Gary Marcus takes the opposite position. He has been one of the most prominent and persistent critics of the idea that pure deep learning can achieve AGI. In his book **“Rebooting AI”** (co-authored with Ernest Davis), Marcus argues that AI systems need built-in knowledge about the structure of the world — concepts like objects, causality, time, and space — that cannot be reliably learned from data alone.

Marcus draws an analogy to child development. Human babies are not blank slates — they come into the world with innate expectations about how physical objects behave (objects persist when hidden, unsupported objects fall, solid objects cannot pass through each other). These are not learned from data; they are built into the brain’s architecture. Marcus argues that AI systems similarly need built-in “core knowledge” about the physical and logical structure of the world, which can then be refined by experience.

Marcus points to the persistent failures of large language models — their inability to reliably count, reason about physical causality, or distinguish truth from plausible fiction — as evidence that something fundamental is missing. His prescription: hybrid systems that combine the perceptual power of neural networks with explicit, structured knowledge representations. He draws an explicit parallel to the human brain, which comes “pre-wired” with innate knowledge structures (as Chomsky argued for language) that are then refined by experience.

10.5 Open Questions

Several fundamental questions remain unresolved in the quest for neuro-symbolic AGI:

Can we fully automate concept learning? Can a system discover its own symbolic concepts entirely from data, without any human guidance? DreamCoder takes steps in this direction, but current systems still require carefully designed training environments.

Is there always a role for human-provided symbolic structure? Or can symbolic reasoning emerge purely from neural learning at sufficient scale? This is the core debate between Marcus and LeCun, and it may take a decade or more to resolve.

How do we handle the scalability challenge? Symbolic reasoning systems become computationally expensive as the number of rules grows. How do we combine the scalability of neural networks with the precision of symbolic reasoning without creating a bottleneck?

What is the right interface between neural and symbolic components? How should symbols be represented so that neural networks can produce them and symbolic systems can consume them? This “neural-symbolic interface” problem is one of the most practical challenges in the field.

? KNOWLEDGE CHECK 2

1. What does Global Workspace Theory propose about consciousness?

- a) Consciousness arises when information is broadcast to multiple specialized processing modules via a central workspace.
- b) Consciousness is located in a single brain region called the global workspace.
- c) Consciousness requires quantum computing processes in neurons.

2. How does AlphaFold combine neural and symbolic approaches?

- a) It uses only neural networks trained on very large datasets of protein structures.
- b) It uses neural networks to learn patterns from known structures AND applies symbolic physics/chemistry rules as constraints.
- c) It uses only symbolic chemistry rules to predict protein folding.

3. What is Gary Marcus's main argument for neuro-symbolic AI?

- a) Neural networks are fundamentally flawed and should be replaced entirely by symbolic AI.
- b) Pure deep learning cannot achieve AGI because it lacks built-in knowledge about the structure of the world — concepts like objects, causality, and time.
- c) Symbolic AI was proven to fail in the 1980s and should be abandoned.

💡 CAREER CONNECTIONS

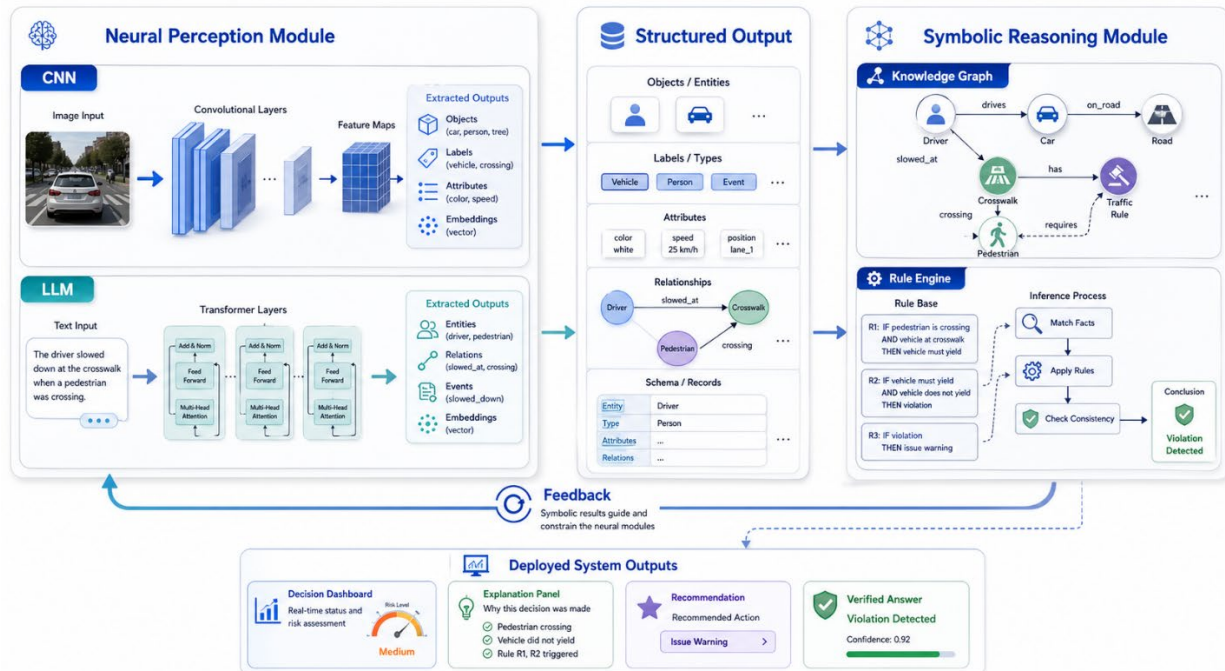
Neuro-symbolic AI is one of the fastest-growing areas in AI research and industry. Here are some places where this expertise is in demand:

- IBM Research — pioneering neuro-symbolic AI through their Neurosymbolic AI initiative, combining knowledge graphs with deep learning for enterprise applications.
- MIT CSAIL — home of DreamCoder and other cutting-edge concept learning research.
- Google DeepMind — developing systems like AlphaFold and Gemini that combine neural perception with structured reasoning.
- Startups — companies like Relation Therapeutics (drug discovery), Symbolica AI (mathematical reasoning), and numerous others are building products on neuro-symbolic foundations.
- Roles to explore: AI Research Scientist, Knowledge Engineer, Ontology Developer, AI Safety Engineer, Machine Learning Engineer with symbolic AI specialization.

PART 4

Lab 10: Neuro-Symbolic Logic Puzzle Solver

Combining Python (Neural) and Wolfram Language (Symbolic)



? KNOWLEDGE CHECK 4

1. Which of the following is the strongest reason that neuro-symbolic integration is considered a promising path toward Artificial General Intelligence?

- a) It uses more computing power than any other approach
- b) It combines the strengths of neural perception and symbolic reasoning — the two capabilities the human brain clearly integrates
- c) It replaces all previous approaches to AI
- d) It requires no training data

2. A bank needs to deploy a loan-approval AI system. Regulators require that every decision be fully explainable. Which architecture fits this requirement best?

- a) A pure deep neural network, because deep networks are the most accurate
- b) A pure symbolic rule system, because rules are the only way to be fair
- c) A neuro-symbolic system that uses neural components for document understanding and symbolic components for the final approval logic
- d) A large language model with no retrieval or rule layer

3. In the context of Retrieval-Augmented Generation (RAG), which part of the system is neural and which part is symbolic?

- a) Both the retriever and the generator are neural
- b) The retrieval over a structured knowledge base is the symbolic part; the language generation is neural
- c) Both parts are symbolic
- d) RAG has nothing to do with neuro-symbolic AI

Answers available from your instructor.

Section Lab

Neuro-Symbolic Logic Puzzle Solver

LAB LEARNING OBJECTIVES

By the end of this lab, you will be able to:

1. Build a simple image classifier in Python (using TensorFlow/Keras) that converts visual inputs into symbolic labels.
2. Encode logic puzzle constraints as symbolic rules in the Wolfram Language using `Solve[]`.
3. Integrate the two components by passing the Python classifier's output into Wolfram as input.
4. Trace the reasoning steps of the combined system to explain how it arrived at its solution.
5. Evaluate the strengths of each component and explain why the hybrid approach outperforms either component alone.

Background

In this lab, you will build a **neuro-symbolic system** that solves a classic logic puzzle — sometimes called a “Zebra puzzle” or “Einstein’s riddle.” These puzzles present a set of clues and ask you to figure out the complete solution using logical deduction. For example: “The person who lives in the red house drinks coffee. The person who owns a dog lives next to the person in the blue house.” From a set of such clues, you must determine who lives where, owns what pet, and drinks which beverage.

What makes our approach special is that instead of typing the clues in as text, we will use visual icons — colored shapes representing houses, pets, and beverages. A Python neural network will classify these icons (the perception step), and the Wolfram Language will solve the logic puzzle from the classified labels (the reasoning step). This mirrors the real-world neuro-symbolic pattern: neural perception feeding into symbolic reasoning.

Architecture Overview

The system has two main components connected by a data pipeline:

Component 1 — Python/TensorFlow Neural Classifier: A convolutional neural network (CNN) trained to recognize simple visual icons. Given an image of a colored shape (e.g., a red square, a blue circle, a green triangle), it outputs a symbolic label like “red_house” or “blue_circle.” This is the perception layer — it converts raw visual data into symbols that the reasoning layer can work with.

Component 2 — Wolfram Language Constraint Solver: A symbolic reasoning system that takes the classified labels and a set of puzzle constraints, encodes them as a system of equations, and uses Wolfram's `Solve[]` function to find the solution. This is the reasoning layer — it applies logical rules to the symbols produced by the neural classifier.

The Bridge: Python saves the classifier's output as a structured data file (JSON or CSV). The Wolfram Language reads this file and uses the labels as inputs to its constraint solver. The key insight: neither component could solve the puzzle alone. The neural network can see the icons but cannot reason about the rules. The symbolic solver can reason about the rules but cannot see the icons.



Part A: The Puzzle Setup

Here is our simplified logic puzzle. There are three houses in a row, each with a different color, a different pet, and a different beverage:

- Colors: Red, Blue, Green
- Pets: Cat, Dog, Fish
- Beverages: Coffee, Tea, Juice

The clues (which will be presented as visual icons):

1. The person in the Red house drinks Coffee.
2. The person who owns the Dog lives in the middle house (House 2).
3. The Green house is to the right of the Blue house.
4. The Tea drinker owns the Cat.



5. The Juice drinker lives in the Green house.

Take a moment to try solving this puzzle yourself before continuing. Working through it by hand will help you appreciate what the symbolic solver does and why logical constraint propagation is so powerful.

Part B: Python Component — The Neural Classifier

In this step, you will create a simple CNN that classifies icon images into symbolic labels. For this lab, we provide a small pre-generated dataset of 9 icon types (3 colors $\times$ 3 categories). The model architecture is straightforward:

1. Load the icon dataset (provided as PNG images organized by label folder).
2. Build a small CNN with two convolutional layers, max pooling, and a dense output layer with softmax activation.
3. Train the model for 10 epochs (this is a simple classification task, so training is fast).
4. Test the model on the puzzle's visual clues — icons representing each house's color, pet, and beverage.
5. Save the predicted labels to a JSON file: {"house_1_color": "red", "house_1_pet": "cat", ...}

The key takeaway from this step: the neural network is doing pure **perception**. It converts pixels into symbols. It has no idea what the puzzle is, what the rules are, or how to solve it. It just classifies images. This is exactly what your visual cortex does when you look at the clue icons — you recognize “red house” before you start reasoning about the puzzle.

Part C: Wolfram Language Component — The Constraint Solver

Now we switch to the Wolfram Language to encode the puzzle's logical constraints. The Wolfram Language has powerful built-in support for symbolic computation and constraint solving. Here is the approach:

1. Define variables for each house: `color[1]`, `color[2]`, `color[3]`, `pet[1]`, `pet[2]`, `pet[3]`, `drink[1]`, `drink[2]`, `drink[3]`.
2. Encode the basic constraints: each house has a unique color, pet, and beverage (all-different constraints).
3. Encode the clue constraints using Wolfram's symbolic syntax:
 - `color[i] == "Red"` implies `drink[i] == "Coffee"`
 - `pet[2] == "Dog"`
 - The Green house's position = Blue house's position + 1
 - `drink[i] == "Tea"` implies `pet[i] == "Cat"`
 - `color[i] == "Green"` implies `drink[i] == "Juice"`
4. Use `Solve[]` or `Reduce[]` to find the unique solution.
5. Output the solution in a readable table format.

The key takeaway: the symbolic solver is doing pure **reasoning**. It applies logical rules to symbols. It cannot perceive the real world — it needs the neural classifier to provide the symbolic inputs. This is like your prefrontal cortex — it can reason logically about “red house” and “coffee” but it relies on your visual system to identify those things in the first place.

Part D: Integration — Connecting Python to Wolfram

The integration step is where neuro-symbolic magic happens. The workflow is:

1. Python classifies all visual icons and saves the results to a JSON file.
2. A bridge script calls the Wolfram Language kernel with the JSON file as input.
3. Wolfram reads the JSON, maps the neural classifier's labels to its symbolic variables, applies the constraints, and solves the puzzle.
4. Wolfram outputs the solution, which the bridge script captures and displays.

You can call the Wolfram kernel from Python using the `wolframclient` library, or you can use a simple file-based pipeline where Python writes JSON and a Wolfram script reads it. Both approaches work — the file-based approach is simpler for this lab.

Part E: Run and Interpret

Run the complete pipeline and observe the solution. Then trace the reasoning steps:

1. Which icons did the neural classifier identify? Were there any it was uncertain about? Check the confidence scores for each classification.
2. How did the symbolic solver use these labels? Which constraint eliminated which possibilities? Trace through the solver's logic step by step.
3. Could the neural classifier have solved the puzzle alone? Could the symbolic solver have solved it without the classifier? Why or why not?
4. What happens if the neural classifier makes an error — say, it misclassifies a red icon as green? How does this affect the symbolic solver? Does the solver detect the inconsistency?
5. How does the combined system's behavior compare to how you solved the puzzle by hand?

DISCUSSION QUESTIONS

1. What would happen if you removed the symbolic reasoning component and tried to train a neural network to solve logic puzzles directly from visual clues? What difficulties would you encounter? How many training examples would you need?
2. In what real-world applications would this kind of neural-to-symbolic pipeline be valuable? Think of situations where perception and reasoning need to work together — in medicine, security, manufacturing, or everyday life.
3. How does this lab system compare to how the human brain might solve the same puzzle? Which parts are similar to System 1 and System 2 processing? At what point does the system switch from “seeing” to “thinking”?

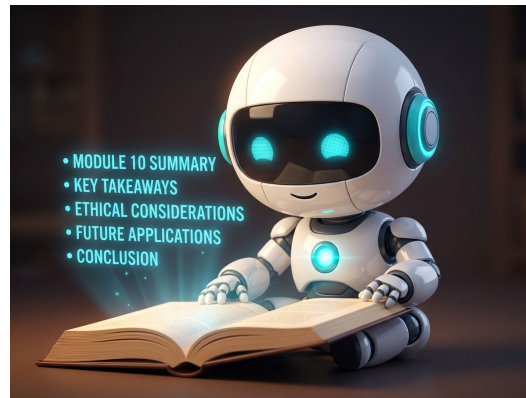
EXTENSION CHALLENGE

Add a sixth clue to the puzzle — for example, “The Fish owner lives in House 1” — and update both the visual clues and the Wolfram constraint set. Run the system again. Does it still find a unique solution? If the new constraint conflicts with existing ones, what does the symbolic solver do? This exercise demonstrates how symbolic systems handle logical consistency — a capability that pure neural networks lack. Try adding a contradictory constraint (one that makes the puzzle unsolvable) and observe how the symbolic solver responds versus how a neural network would respond.

Section

Module 10 Summary

This module explored the frontier of neuro-symbolic AI — the quest to build systems that combine the pattern recognition power of neural networks with the logical reasoning capabilities of symbolic AI. We traced the biological roots of this integration in the human brain and examined real-world architectures that implement it.



Concept	Key Idea
Neural Networks	Excel at perception (image recognition, language processing) through pattern matching, but cannot explain reasoning or apply logical rules to novel situations.
Symbolic AI	Excel at logical reasoning with explicit rules and knowledge representations, but cannot handle real-world perception or situations not covered by their rules.
Neuro-Symbolic AI	Hybrid systems that combine neural perception with symbolic reasoning, aiming to capture the strengths of both approaches while mitigating their weaknesses.
System 1 / System 2	Kahneman's dual-process framework maps to neural (fast, automatic, pattern-matching) and symbolic (slow, deliberate, rule-following) processing.
Concept Learning	The ability to extract abstract, generalizable concepts from few examples — a key capability gap between humans and current AI systems.
Global Workspace Theory	Baars' model of consciousness as a central broadcast mechanism coordinating specialized modules — an influential blueprint for AI architecture.
Knowledge Graphs	Structured representations of entities and relationships that provide symbolic knowledge to neural systems (e.g., Google's Knowledge Graph, RAG systems).
Integration Strategies	Four approaches: Sequential (perceive then reason), Guided (rules constrain training), Compiled (rules into weights), Bidirectional (continuous feedback).
AlphaFold	DeepMind's protein structure prediction system — neural networks guided by symbolic physics/chemistry constraints. Won 2024 Nobel Prize.
Path to AGI	Leading researchers argue that AGI requires neuro-symbolic integration, not just scaling neural networks larger.
Active Inference	From Module 09: the brain's prediction-correction loop can be seen as a neuro-symbolic process — neural predictions constrained by a symbolic world model.

Five Key Takeaways

KEY TAKEAWAY

Neural networks are extraordinary pattern recognizers but poor logicians. They learn from data, excel at perception, and achieve superhuman performance on many tasks — but they cannot explain their reasoning, apply rules to novel situations, or reliably avoid confident errors.

KEY TAKEAWAY

Symbolic AI provides transparency, explainability, and rigorous reasoning — but it cannot perceive the messy real world or handle situations its rules do not cover. The knowledge acquisition bottleneck makes it impractical to manually encode everything.

KEY TAKEAWAY

The human brain is the proof of concept for neuro-symbolic integration. System 1 (neural, fast, intuitive) and System 2 (symbolic, slow, deliberate) work together seamlessly, with each system informing and constraining the other.

KEY TAKEAWAY

Global Workspace Theory provides an architectural blueprint: specialized neural modules communicating through a shared symbolic workspace, coordinated by attention and broadcasting mechanisms. Modern AI agent frameworks are approximate implementations of this idea.

KEY TAKEAWAY

The path to Artificial General Intelligence likely runs through neuro-symbolic integration. Whether through explicit hybrid architectures or emergent symbolic reasoning in large neural systems, the field is converging on the insight that both perception and reasoning are essential for general intelligence.

 REQUIRED READING

Marcus, G. & Davis, E. (2019). *Rebooting AI: Building Artificial Intelligence We Can Trust*. New York: Pantheon Books. An accessible, widely read book making the case for hybrid neuro-symbolic approaches — directly relevant to Sections 2, 3, and 10. No technical background required.

 LOOKING AHEAD — MODULE 11

In Module 11, we will explore Embodied Cognition and Robotics — the idea that true intelligence requires a body that interacts with the physical world. We will see how the neuro-symbolic architectures from this module become even more powerful when connected to sensors and actuators that engage with real environments. Can a disembodied AI ever truly understand the world? Does a robot need to touch, push, and manipulate objects to develop genuine concepts? Module 11 investigates.

 KNOWLEDGE CHECK 5

1. Which combination of responsibilities makes the most sense in a neuro-symbolic medical imaging system?

- a) The neural network handles the final medical diagnosis; the symbolic system handles image pixels
- b) The neural network identifies features in the image; the symbolic system applies clinical guidelines and explains the diagnosis
- c) Both the neural and symbolic components do the same work in parallel
- d) Only the symbolic component is used; the neural network is optional

2. Which of the following statements about large language models is most accurate, based on this module?

- a) LLMs are pure symbolic reasoners
- b) LLMs show impressive System 1-like behavior, but often struggle with tasks requiring explicit multi-step logic
- c) LLMs never make factual errors
- d) LLMs have proven that symbolic AI is no longer needed

3. Which approach is most likely to reduce LLM hallucinations in a customer-service chatbot?

- a) Making the language model larger
- b) Adding a retrieval step that grounds the model's responses in a verified knowledge base
- c) Removing all structured data from the system
- d) Training the model on more creative text

Answers available from your instructor.

SECTION 11

Review Questions

Work through these questions to reinforce and extend what you learned in this module. Questions 1 through 5 test your grasp of core concepts. Questions 6 through 10 ask you to apply neuro-symbolic thinking to real scenarios you might encounter in industry.

Conceptual Understanding

1. In your own words, explain why a neural network that has learned to recognize 100,000 images of cats can still fail on a cat wearing a costume, while a child who has seen only a handful of cats rarely makes that mistake. What is the missing ingredient, and which branch of AI historically provided it?
2. A hospital wants to build an AI system that looks at chest X-rays and reports whether a patient has pneumonia. Describe how a pure neural approach, a pure symbolic approach, and a neuro-symbolic approach would each handle this task. Which would you recommend and why?
3. System 1 and System 2 thinking, as described by Daniel Kahneman, are often used as a metaphor for neuro-symbolic AI. Explain the metaphor. Then describe one task where System 1 alone would fail and one task where System 2 alone would fail.
4. What does Global Workspace Theory propose about how the brain integrates information from specialized regions? How does this biological architecture inspire the design of modern neuro-symbolic systems?
5. Retrieval-Augmented Generation (RAG) is sometimes described as a "lightweight neuro-symbolic technique." Explain what this means. Which part is neural? Which part is symbolic? Why does this combination reduce hallucinations in large language models?

Applied Scenarios

6. You are a new hire at a bank. Your team builds an AI system that decides whether to approve or deny loan applications. Regulators require that every decision be fully explainable to the applicant. Design a neuro-symbolic architecture that meets this requirement. Which responsibilities would you assign to the neural component and which to the symbolic component?
7. A self-driving car must decide whether to proceed through an intersection where a pedestrian is standing on the curb. List three perception tasks (neural) and three reasoning tasks (symbolic) involved in this decision. Explain how the two components would need to exchange information in real time.
8. Your company is building a customer service chatbot powered by a large language model. The chatbot keeps "making up" policies that do not exist in the actual company handbook. Propose a neuro-symbolic fix. Specifically, describe the role of a knowledge graph or structured policy database in your solution.
9. Imagine you are asked to explain neuro-symbolic AI to a group of high school students who have never heard of it. Create a short analogy (not the ones used in this booklet) that captures the core idea of combining pattern recognition with logical reasoning.
10. The road to Artificial General Intelligence is often described as running through neuro-symbolic integration. Do you agree? Write a short argument either supporting or challenging this claim, drawing on at least two specific examples from this module.

SECTION 12

Glossary

Key terms used throughout Module 10. Refer back to this glossary as you work through the lab and as you read the references listed in Section 13.

Artificial General Intelligence (AGI). A hypothetical form of AI that can perform any intellectual task a human can, transferring knowledge flexibly across domains. Most researchers believe AGI will require neuro-symbolic integration.

Concept Learning. The cognitive process of forming abstract categories from specific examples. The brain builds concepts through hierarchical abstraction, starting from raw perception and ending in rule-like representations.

Connectome. The complete map of neural connections in a brain. Often described as the brain's biological knowledge graph because concepts are represented by patterns of connected regions, not by single neurons.

Constraint Satisfaction. A symbolic reasoning technique where a system finds values for variables that satisfy a set of logical rules or constraints. Used in the Module 10 lab to solve the Zebra-style puzzle.

Dual-Process Theory. The psychological theory, popularized by Daniel Kahneman, that human thought uses two distinct systems: System 1 (fast, intuitive, pattern-based) and System 2 (slow, deliberate, rule-based).

Emergent Abilities. Capabilities that appear in large language models at scale but are not present in smaller versions. Whether these reflect genuine concept learning or sophisticated pattern matching is an open research question.

Explainability. The property of an AI system whose decisions can be understood and traced by humans. Symbolic systems are naturally explainable; neural systems often are not. Neuro-symbolic systems aim to combine performance with explainability.

Global Workspace Theory (GWT). A theory of consciousness proposed by Bernard Baars stating that specialized brain modules compete to broadcast information on a shared workspace. GWT serves as an architectural blueprint for modern neuro-symbolic AI.

Hallucination (in LLMs). When a large language model generates confident but factually incorrect statements. Hallucinations often stem from the model having no external source of truth. Knowledge graphs and retrieval systems are used to reduce them.

Hybrid AI. An umbrella term for systems that combine two or more AI paradigms, most commonly neural networks with symbolic reasoning. Neuro-symbolic AI is the most active form of hybrid AI today.

Knowledge Graph. A structured representation of facts as nodes (entities) and edges (relationships). Used in neuro-symbolic systems to give neural models access to verifiable, structured knowledge.

Large Language Model (LLM). A neural network trained on massive text corpora that can generate, summarize, and reason about natural language. Modern LLMs like GPT-4 show impressive System 1-like behavior but struggle with tasks requiring explicit logic.

Neural Network. A computing system inspired by biological neurons, made of layers of connected nodes that learn patterns from data. Excels at perception and pattern recognition but not at explicit reasoning.

Neuro-Symbolic AI. A research area that combines neural pattern recognition with symbolic logical reasoning in a single system. The goal is to capture the strengths of both while compensating for their weaknesses.

Perception. The process of interpreting raw sensory input into meaningful information. Neural networks, like the human visual system, are specialists in perception.

Reasoning. The process of drawing conclusions by applying logical rules to known information. Historically the domain of symbolic AI.

Retrieval-Augmented Generation (RAG). A technique that augments a neural language model with access to an external knowledge base. The model retrieves relevant facts before generating a response. A practical, deployed form of neuro-symbolic AI.

Symbolic AI. The branch of AI that represents knowledge using symbols and manipulates them using explicit logical rules. Transparent and rigorous, but unable to handle messy real-world perception on its own.

System 1. In dual-process theory, the fast, automatic, intuitive mode of thinking. The neural-pattern-recognition half of the brain.

System 2. In dual-process theory, the slow, deliberate, rule-following mode of thinking. The symbolic-reasoning half of the brain.

Wolfram Language. A symbolic programming language designed for mathematical, logical, and knowledge-based computation. Used in the Module 10 lab as the symbolic solver component.

Transparency. The degree to which an AI system's internal decision-making can be inspected and understood. Symbolic systems offer high transparency; deep neural networks offer very little. Neuro-symbolic systems aim for transparency where it matters most, such as in regulated industries.

SECTION 13

References and Further Reading

The following sources informed this module and are recommended for students who want to go deeper. The foundational readings are accessible to undergraduates; the research papers are more technical and best approached after completing the module.

Foundational Readings

Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux. The definitive popular treatment of System 1 and System 2 thinking.

Marcus, G., & Davis, E. (2019). *Rebooting AI: Building Artificial Intelligence We Can Trust*. Pantheon Books. A highly readable argument for neuro-symbolic AI, written for a general audience.

Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson. The standard textbook; see chapters on knowledge representation, logical reasoning, and learning.

Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press. The original formulation of Global Workspace Theory.

Hawkins, J. (2021). *A Thousand Brains: A New Theory of Intelligence*. Basic Books. A perspective on how the neocortex builds reference frames and concepts.

Research Papers and Technical Reports

Garcez, A. d'A., & Lamb, L. C. (2023). Neurosymbolic AI: The 3rd Wave. *Artificial Intelligence Review*, 56(11), 12387-12406. A comprehensive survey framing neuro-symbolic AI as the next major wave of AI research.

Mao, J., Gan, C., Kohli, P., Tenenbaum, J. B., & Wu, J. (2019). The Neuro-Symbolic Concept Learner: Interpreting Scenes, Words, and Sentences from Natural Supervision. *International Conference on Learning Representations (ICLR)*. A foundational paper demonstrating how a neuro-symbolic system can learn visual concepts.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474. The original RAG paper.

Marcus, G. (2020). The Next Decade in AI: Four Steps Towards Robust Artificial Intelligence. *arXiv preprint arXiv:2002.06177*. Argues that hybrid systems are essential for progress beyond pure deep learning.

Bengio, Y. (2017). The Consciousness Prior. *arXiv preprint arXiv:1709.08568*. Proposes a neural architecture inspired by Global Workspace Theory.

Dehaene, S., Lau, H., & Kouider, S. (2017). What is Consciousness, and Could Machines Have It? *Science*, 358(6362), 486-492. A readable neuroscience perspective on machine consciousness and global workspace dynamics.

Kautz, H. (2022). The Third AI Summer: AAAI Robert S. Engelmore Memorial Lecture. *AI Magazine*, 43(1), 105-125. A historical and forward-looking overview of neuro-symbolic integration.

Tools and Documentation (for the Lab)

Wolfram Research. (2026). Wolfram Language Documentation. <https://reference.wolfram.com/language/>. Reference for the symbolic component of the lab.

Wolfram Research. (2026). WolframClient for Python. <https://reference.wolfram.com/language/WolframClientForPython/>. Documentation for the Python-to-Wolfram bridge.

PyTorch Team. (2026). PyTorch Documentation. <https://pytorch.org/docs/>. Reference for the neural classifier component.

LangChain. (2026). LangChain Documentation. <https://python.langchain.com/docs/>. Framework documentation for building RAG pipelines, referenced in the career connections callout.

— End of Module 10 —

ITAI 4374: Neuroscience as a Model for AI
Author: Patricia McManus • Collaborator: Claude (Anthropic)